Chapter 7

Photon Based Lithography-Ⅱ

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Outline

- **7. 1. Basic concepts**
- **7. 2. Exposure systems/Lithographic materials**
- **7. 3. Key Factors in Optical Lithography**
- **7. 4. Resolution limits**
- **7. 5. Resolution enhancement techniques**
- **7. 6. EUV**

Outline

- 7. 1. Basic concepts
- 7. 2. Exposure systems/Lithographic materials
- 7. 3. Key Factors in Optical Lithography
- 7. 4. Resolution Limits
- 7. 5. Resolution Enhancement Techniques
- 7. 6. EUV

Key Factors in Optical Lithography

- Placement accuracy (i.e., overlay) （套刻精度）
- Resist response （响应）
- Resolution （分辨率）
- Depth of Focus （焦深）

Exposure: Basic Process

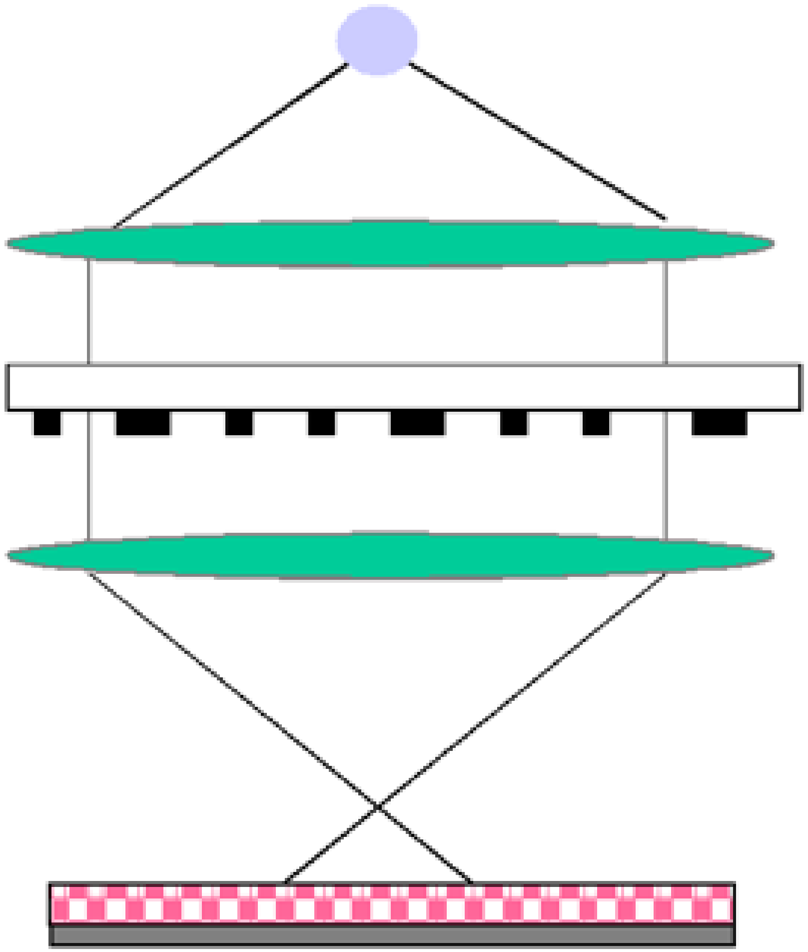
Source

Condenser

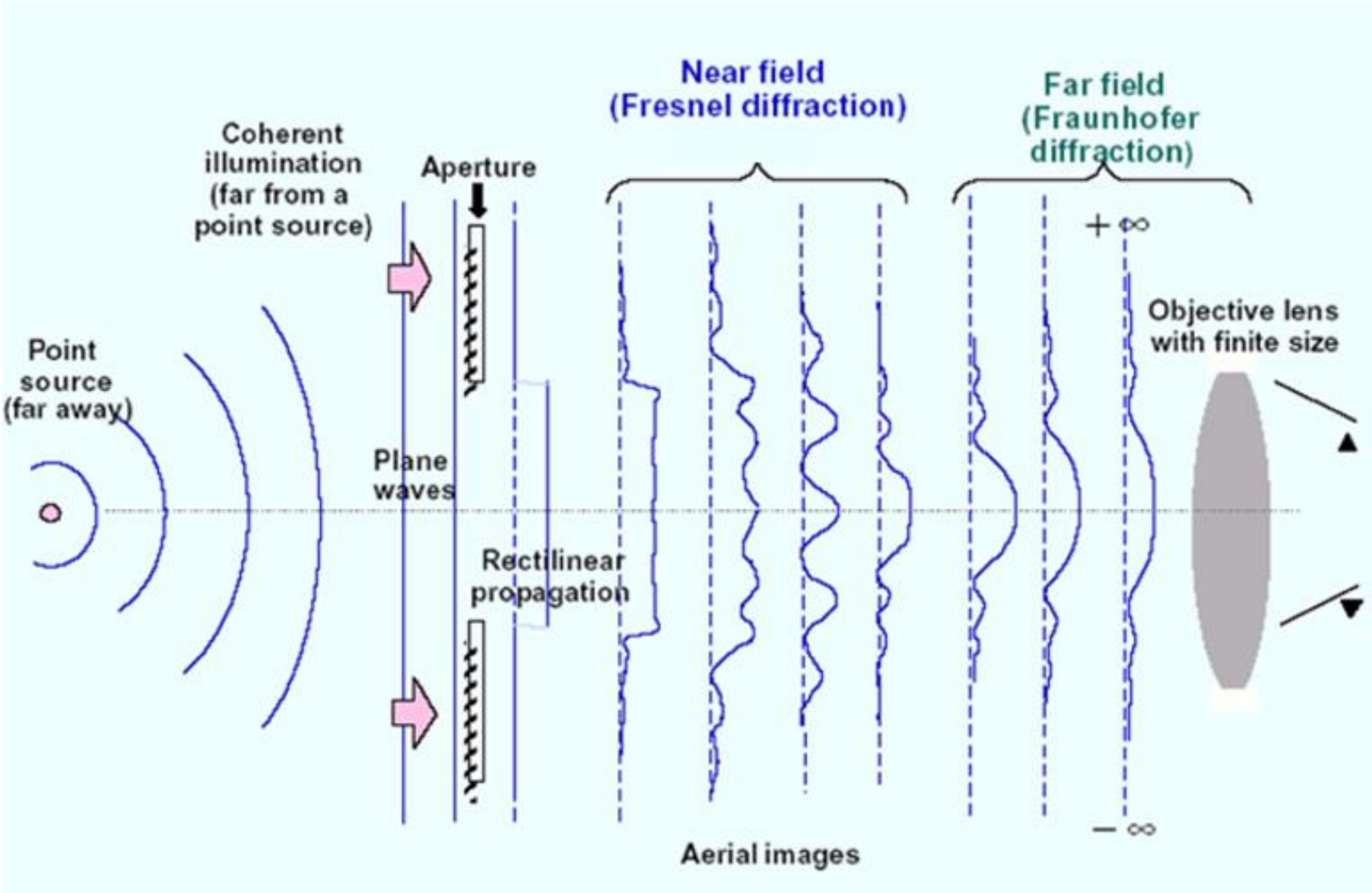
Mask

Reduction optics

Image in resist on sample

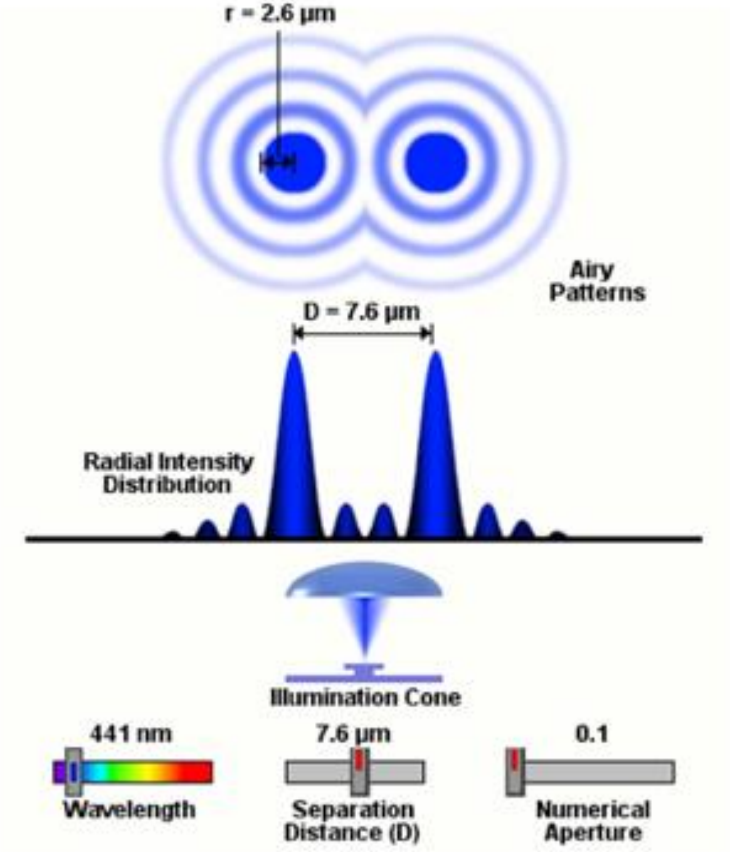


Resolution Limits: Diffraction



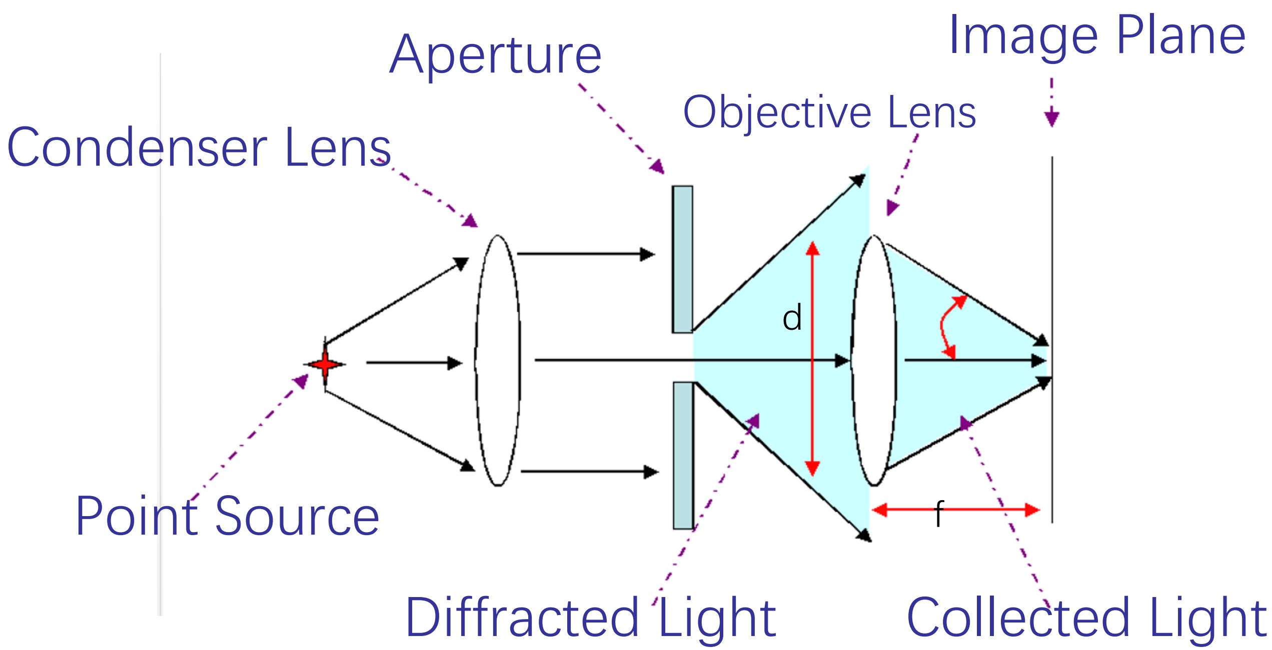
Resolution Limits

Resolution



http://micro.magnet.fsu.edu/primer/java/imageformation/rayleighdisks/

Numerical Aperture



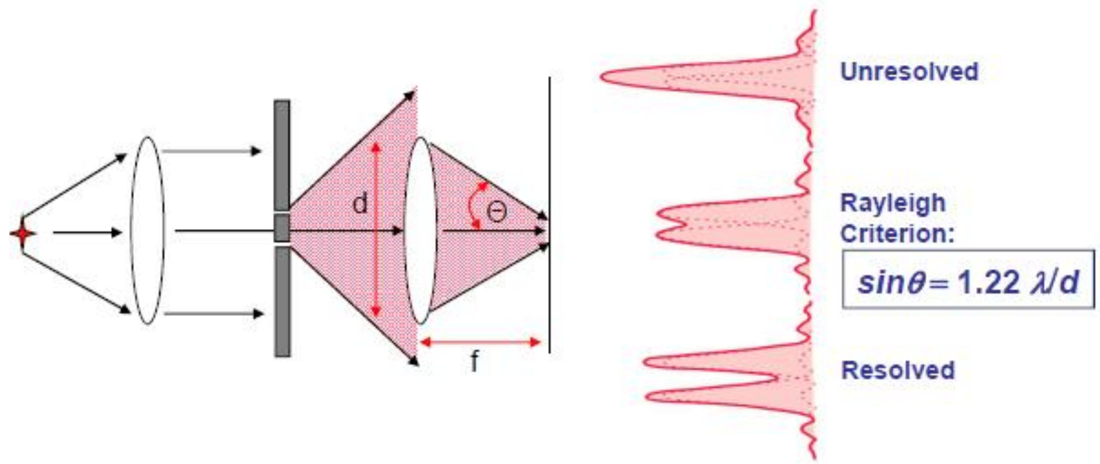
$$d = 2[nf\sin(\theta)]$$

$$NA = n\sin(\theta)$$

$$\sim d_{lens}/f$$

Resolution Limits

Resolution: Rayleigh Criterion



$$R = \frac{1.22 f\lambda}{d} = \frac{1.22 f\lambda}{n(2f\sin\theta)} = \frac{0.61\lambda}{n\sin\theta}$$

Rayleigh Equation

(Circular aperture, far field)

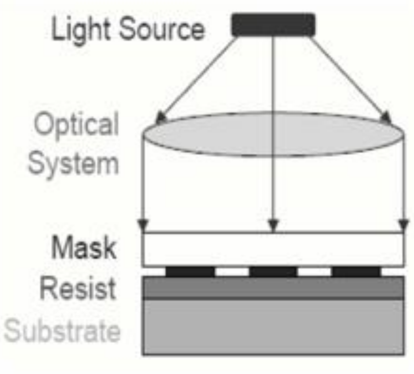
$$\ell_{min} = k_1 \frac{\lambda}{NA}$$

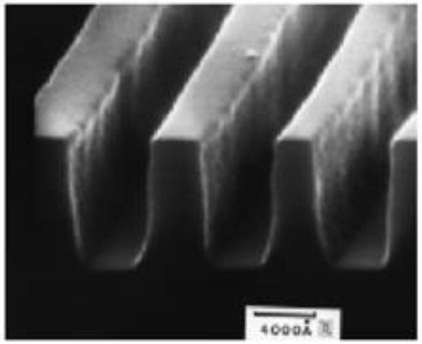
k1 = constant

depends on optics, resist and process latitude

Contact Printing

- 1:1 printing
- No lens
- *Ideal resolution: << 1 μm* (can be very high)
  - **near-field**
  - **→not diffraction limited**
- *Practical resolution: > 1 μm*
- Defects or damage possible





Requirements for optimum resolution:

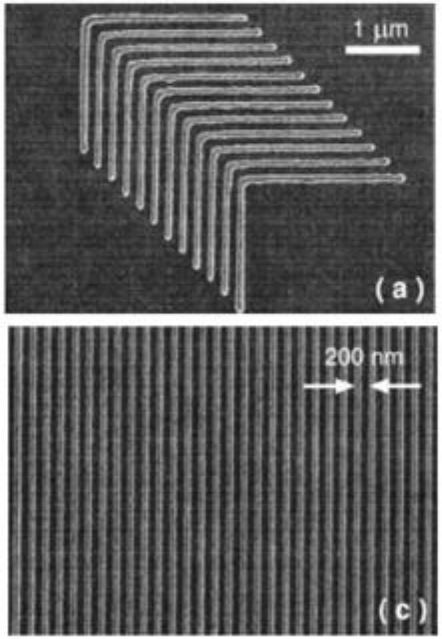
- Optically flat mask & substrate
- Conformable mask
- Conformable substrate

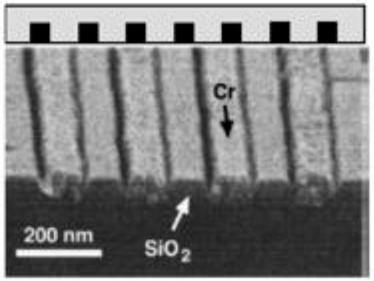
i.e., perfect, intimate contact between mask and substrate

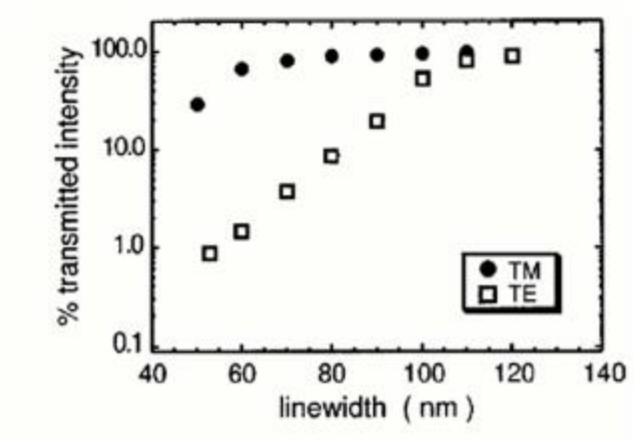
Nanometer-Scale Features by Contact Printing

Conformable embedded-amplitude mask

λ= 220 nm



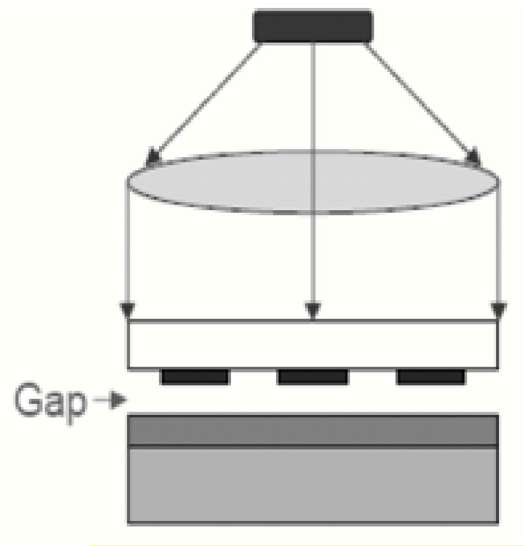




Practical resolution limited by reduction in transmitted intensity

Goodberlet, APL **76**, 667 (2000)

Proximity Printing



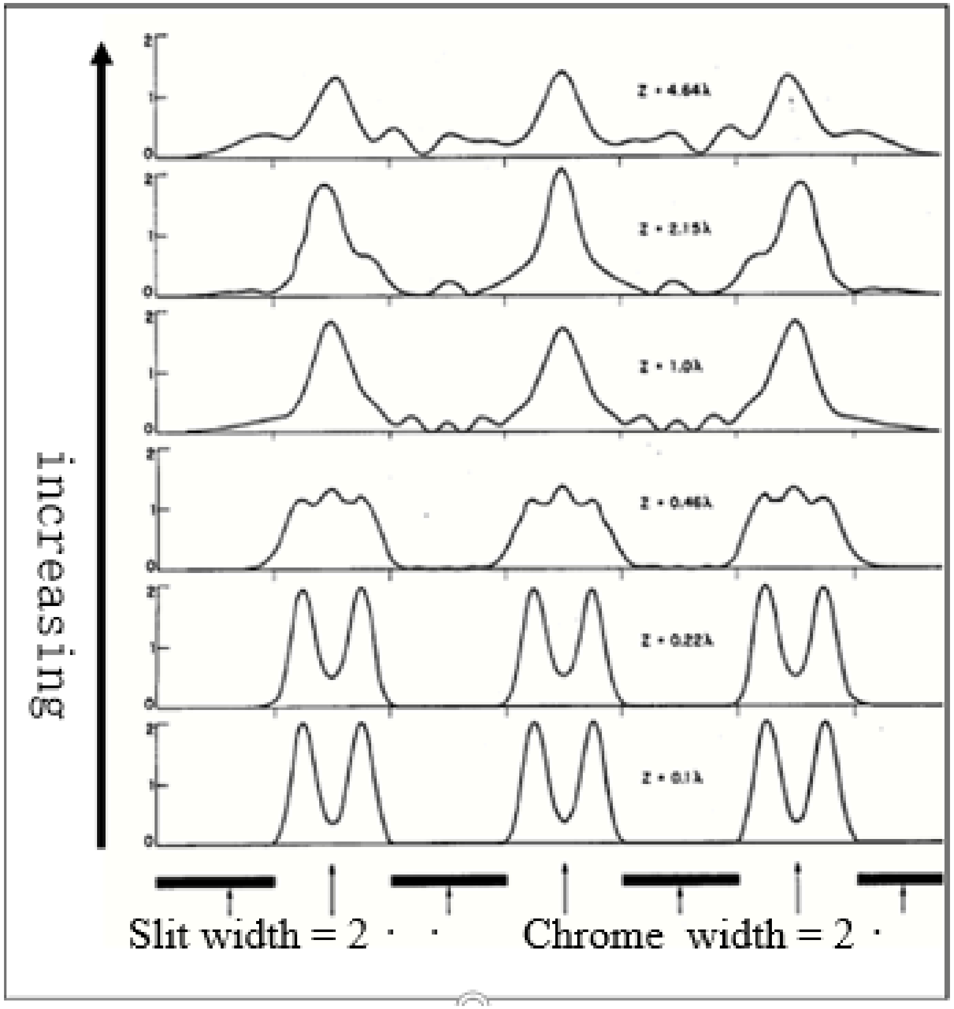
$$l_{min} \sim (\lambda g)^{1/2}$$

~ 2 μm for visible photons

(much shorter for x-rays)

Resolution Limits

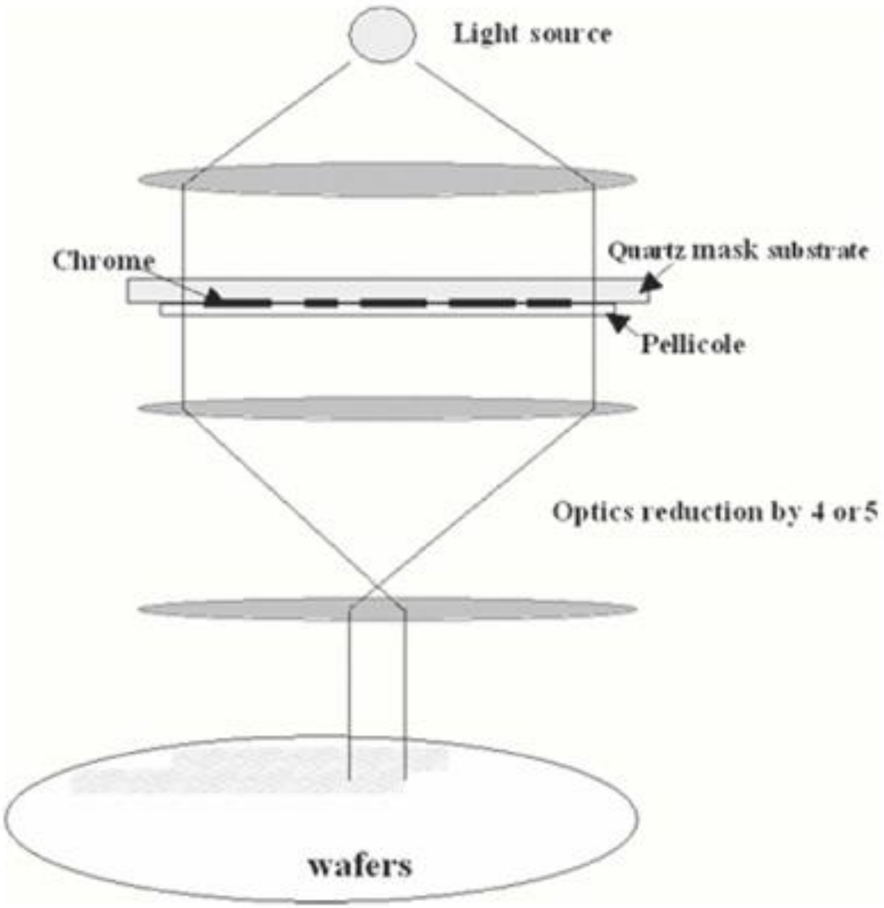
Proximity Printing



Intensity as a function of position on wafer for proximity printing with increasing mask-wafer gap (g)

Resolution Limits

Projection Printing



Demagnification: nX

•10X

•5X

•4X

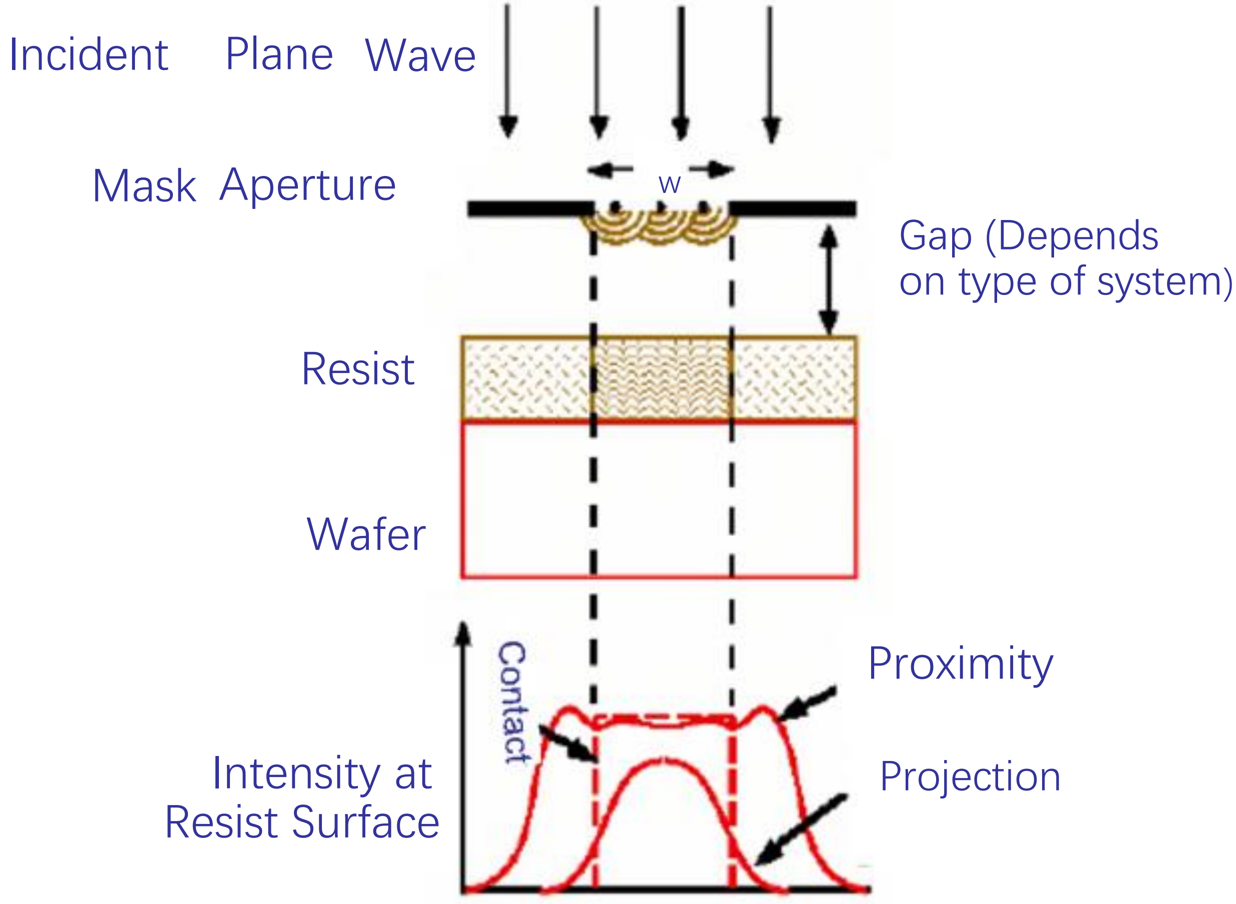
•Very high resolution (deep UV photon

•Complicated optics

•Expensive

AERIAL Image

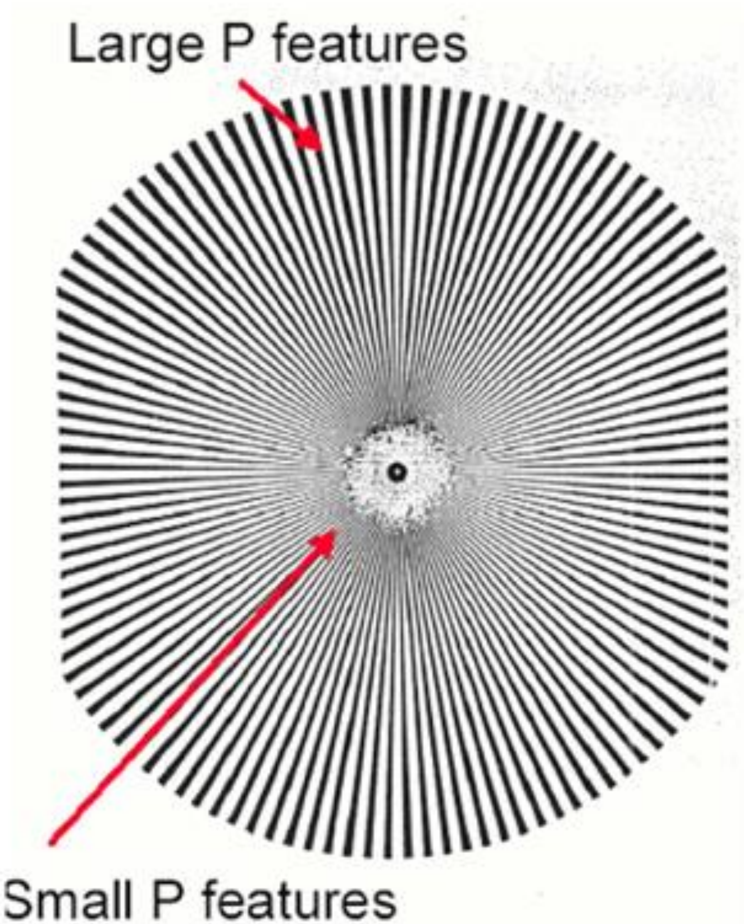
Contact, Proximity, Projection Printing



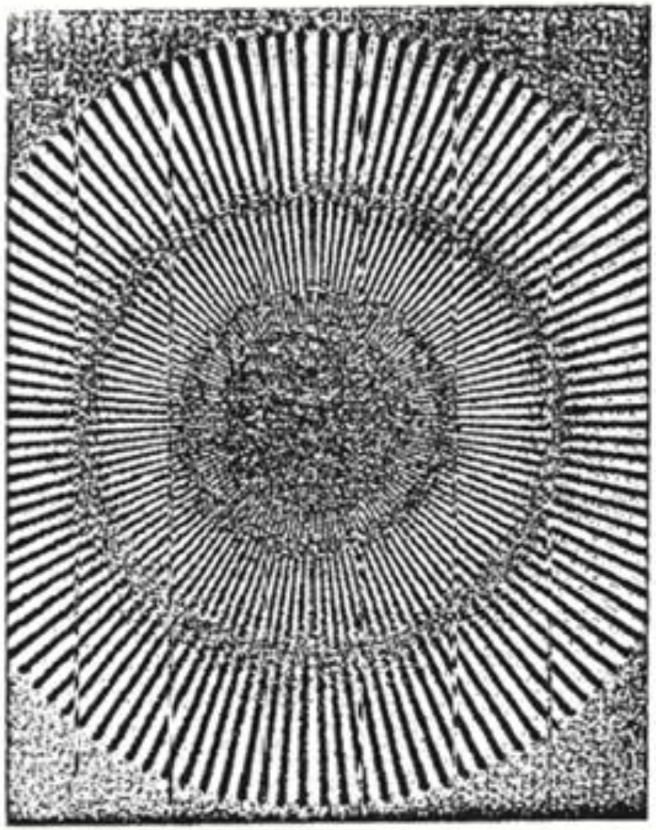
AERIAL:automated enhanced resolution imaging for advanced lithography (illuminator)

Resolution Limits

Focus

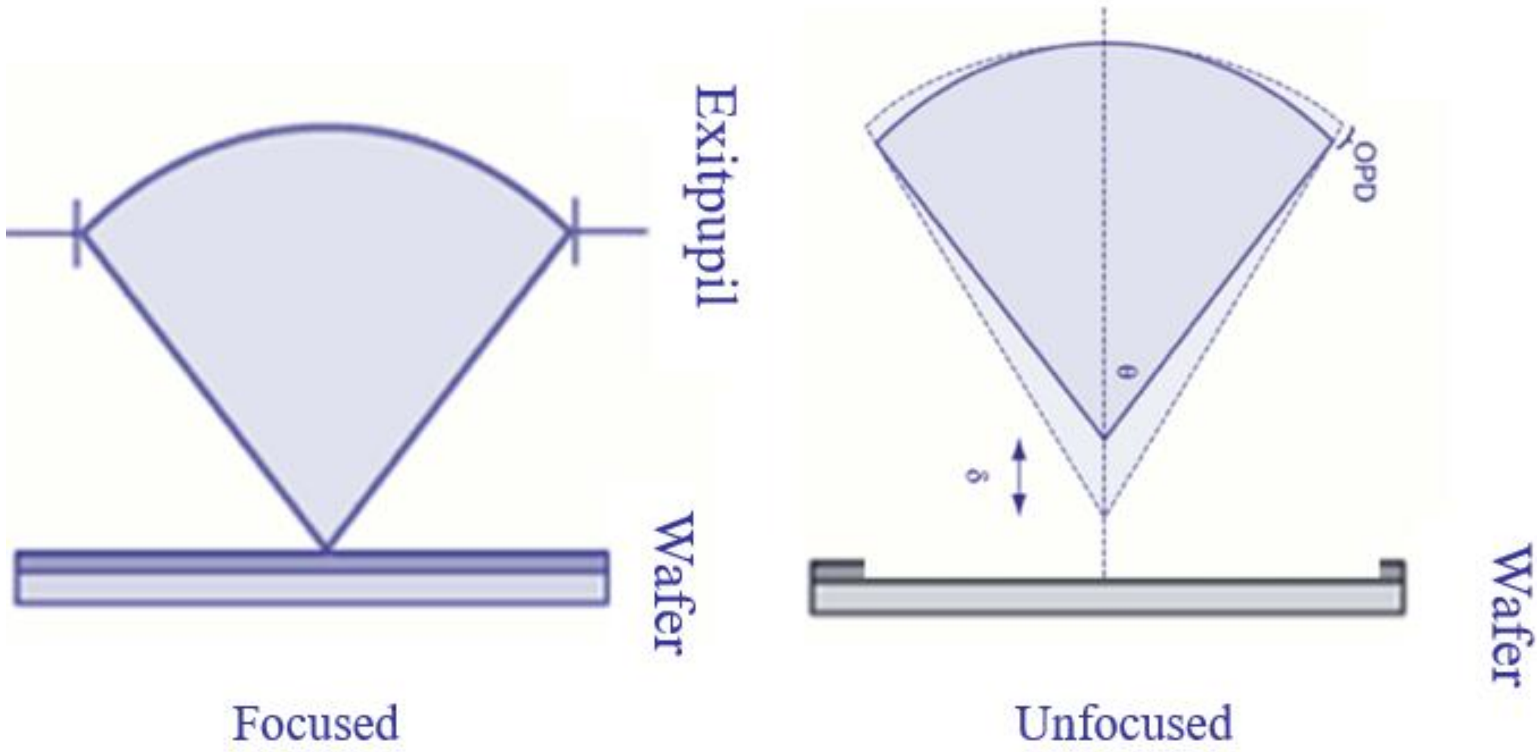


Best focus



Extreme Defocus

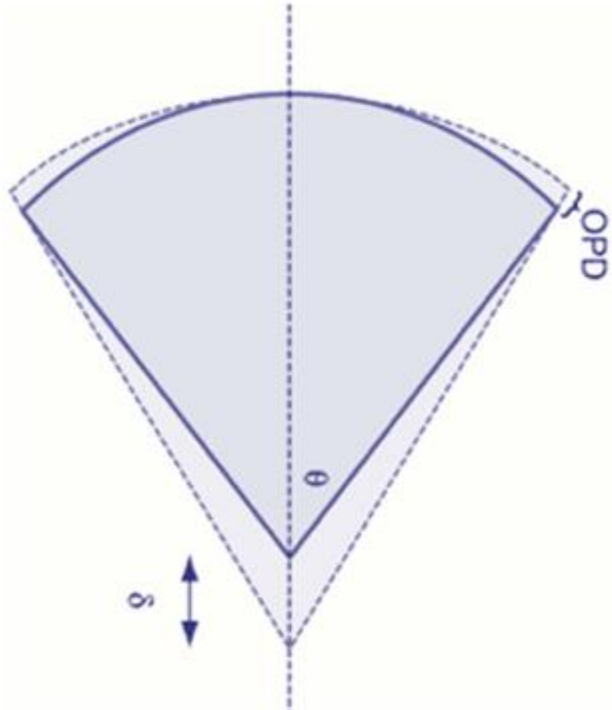
Focus



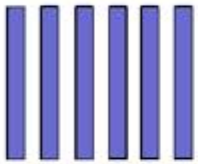
Optical Path Difference $OPD = \delta(1 - \cos\theta)$

Resolution Limits

Focus



$$OPD = \delta(1 - \cos\theta)$$
$$\approx \tfrac{1}{2}\,\delta\sin^2\theta$$



$\sin\theta = m\lambda/p$ for a 1:1 grating

$$OPD_{max} = k_2\ \lambda/4 \quad (k_2 < 1)$$

$$DOF = 2\delta_{max} = \frac{k_2\,\lambda}{\sin^2\theta}$$

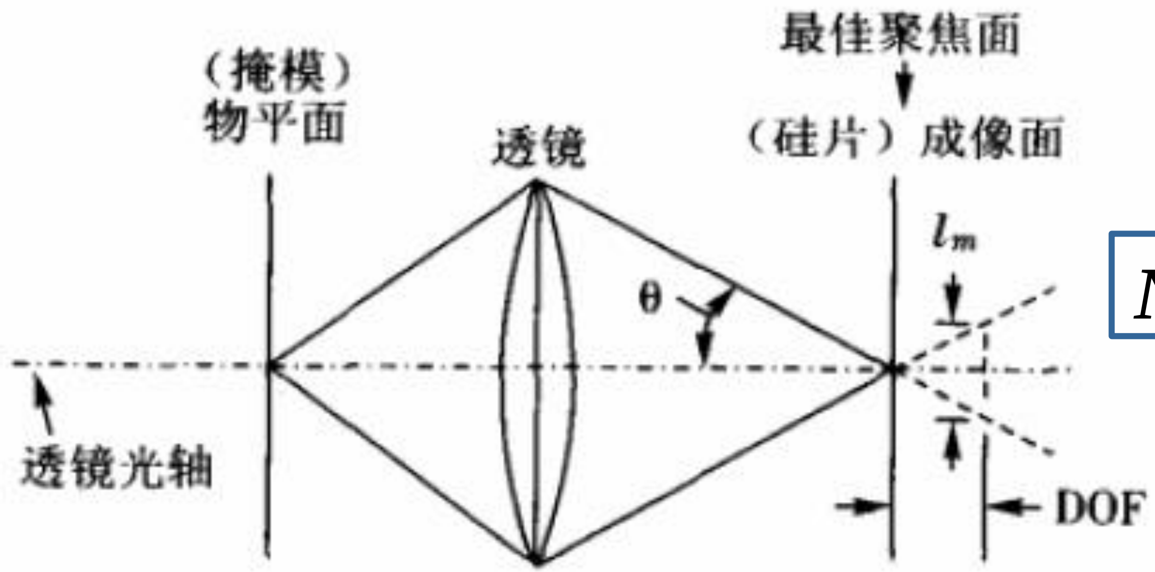
$$= k_2 \frac{\lambda}{(NA)^2}$$

焦距（焦点、焦平面）和焦深的区别?

离开焦面一定范围所成的像是能够满足一定分辨率的要求，这个范围就是我们所说的焦深（DOF）

Resolution Limits

$$DOF = \frac{\pm l_m/2}{\tan^2\theta} \approx \frac{\pm l_m/}{\sin^2\theta} = k_2\frac{\lambda}{(NA)^2}$$



NA↑，焦深↓↓

Resolution vs. DOF

$$\ell_{min} \sim \lambda/NA$$

$$DOF \sim \lambda/2(NA)^2$$

Need: small l_{min}, but large DOF!

Outline

Improving Resolution

Reduce λ

$$l_{min} = k_1\lambda/NA$$

Increase NA

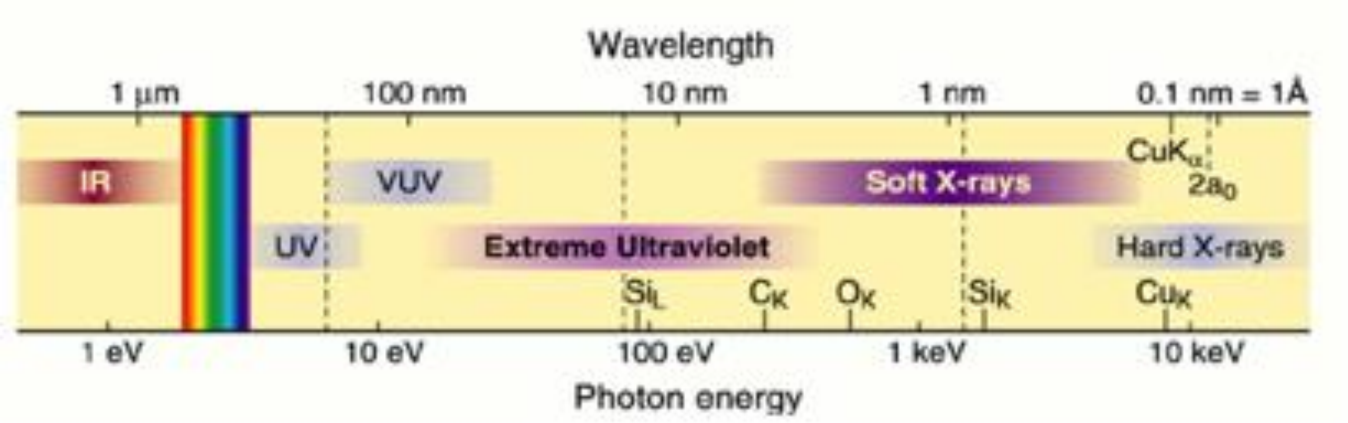
Reduce k_1

k_1 is a function of:

- Optics
  - lens quality (aberrations, etc.)
  - illumination conditions
    - degree of coherence and intensity distribution in the aperture plane
- Mask
- Resist and process latitude

For simplicity, l_{min} is usually taken as P/2, where P is the pitch of a periodic grating

Electromagnetic Spectrum



Shorter λ:
- Microscopy -→ See smaller features
- Lithography -→ Write smaller patterns

(Depending on elemental and chemical sensitivity)

$h\omega\ \lambda = hc = 1239.842$ eV nm

Enhancement Techniques

Reducing λ

Shorter wavelength $l_{min} = k_1\lambda/NA$

Better resolution

Challenges with new sources:

- Optics compatibility
- Narrow bandwidth
- Resist response
  - polymer absorption at λ
- Economics
  - New tooling
  - New materials
- Contamination
  - Energetic photon chemistry

Source	λ
G-line Hg Arc	436nm
I-line Hg Arc	365
KrF excimer	248.3
ArF excimer	193.4
F_2 excimer	157.6
EUV ?laser plasma?	13

% Reduction

16%
32%
22%
19%
91%

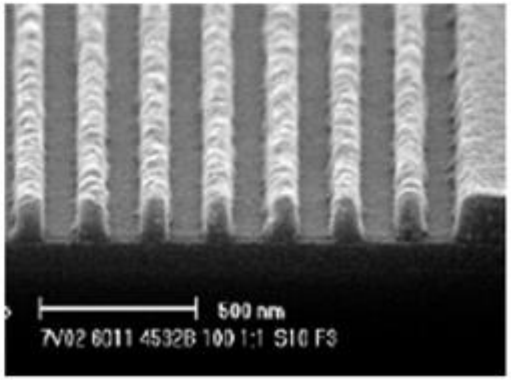
Enhancement Techniques

F2 Lithography

(λ = 157 nm)
Mask: Modified fused silica

Challenges:

- No pellicle (protection for mask)
- Lens material: CaF_2
- Poor optical quality
- Resist materials
- Cost



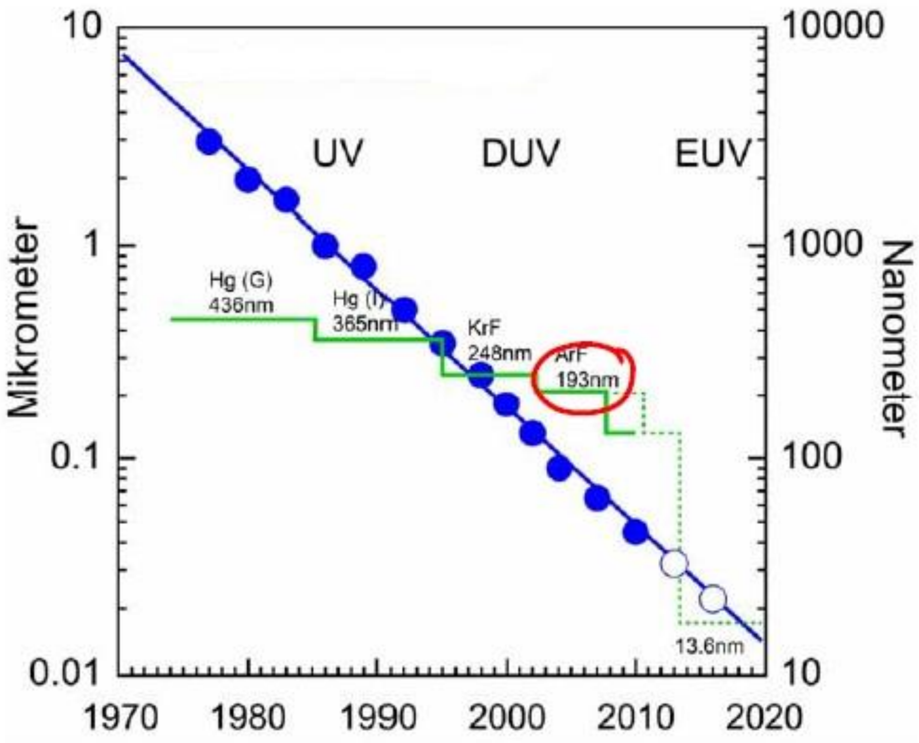
200 nm pitch grating
(C. Robinson et al., SPIE 2004)

	Name	Wavelength (nm)	Application feature size (μm)
Mercury Lamp	G-line	436	0.50
	H-line	405	
	I-line	365	0.35 to 0.25
Excimer Laser	XeF	351	
	XeCl	308	
	KrF (DUV)	248	0.25 to 0.13
	ArF (DUV)	193	0.09 to 0.045
Fluorine Laser	F_2 (VUV)	157	?

2010年，193nm浸入式光刻实现了32nm制程（corei7）
后来又结合双重成像技术，
延申到22nm(2013),14nm(2017)

Enhancement Techniques

Optical Lithography



Since several years: critical dimensions significantly smaller than wavelength of light source

Use resolution enhancement techniques(tricks)

How?

- Off-axis illumination
- Phase shift mask
- Optical proximity correction

Increasing NA

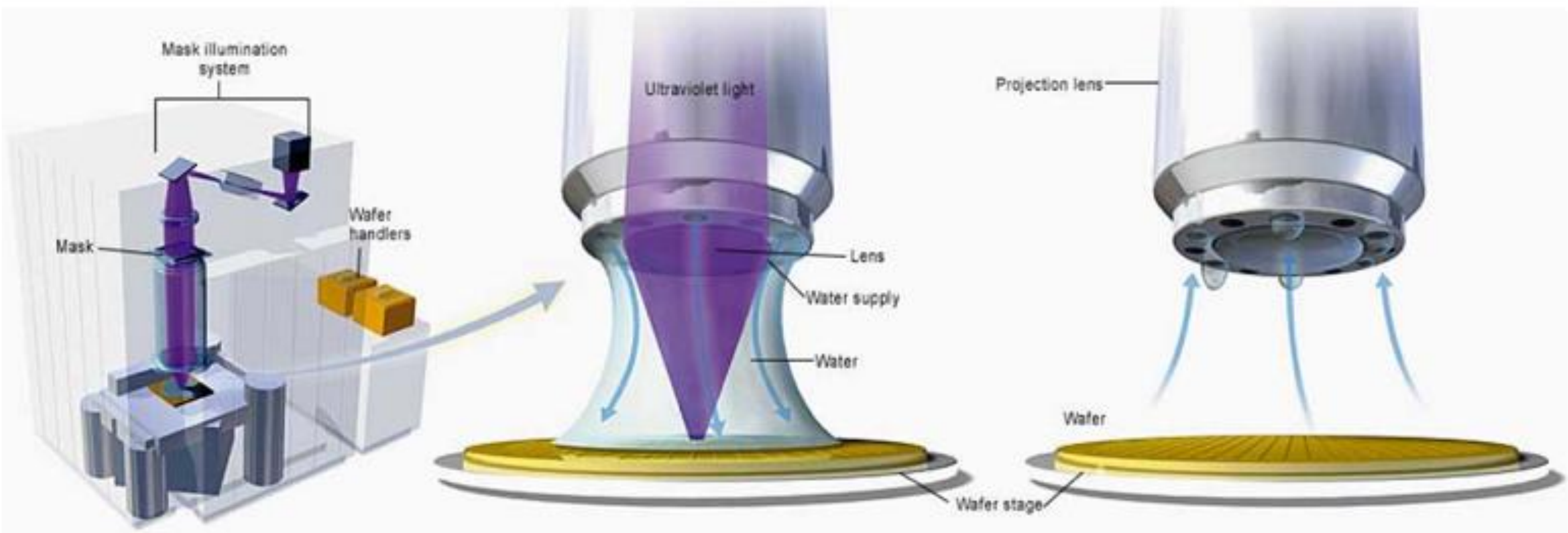
- DOF scaling dominates
  (For NA ≈ 1 , DOF ≈ λ/2)

$$DOF \approx \pm \frac{\lambda}{2(NA)^2}$$

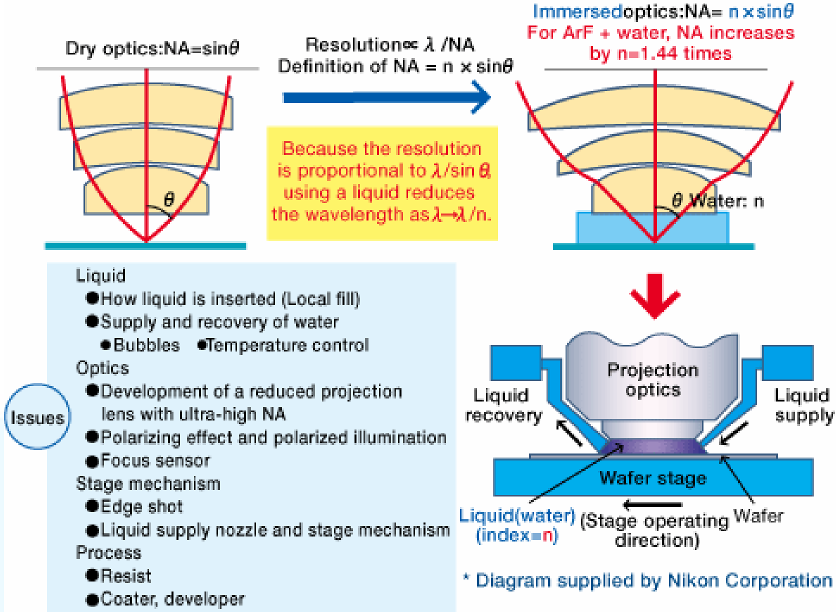
- Great improvements in recent years
  - *Optical technology improved more than expected*
  - *Lens design / fabrication allow NA near 1, (0.93 achieved)*
  - *Advanced processes can reduce focus budget*
  - *Process topography, e.g. CMP, wafer flatness*
  - *Tool control, auto-focus errors*
- Super high NA issues
  - *Small DOF*
  - *Optimized thin resist process*
  - *Polarization and birefringence within optics*

Improving Resolution and DOF:

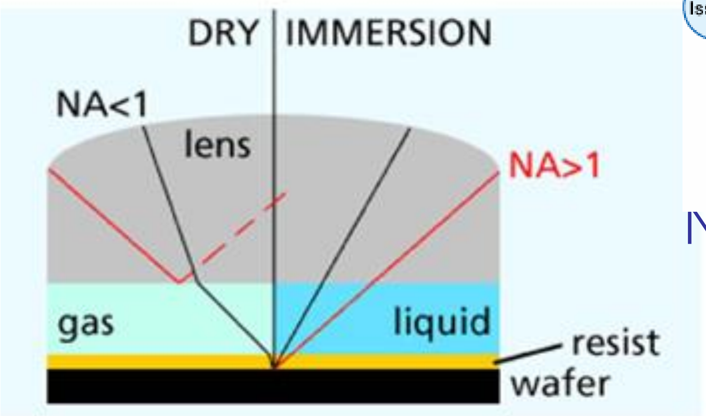
Immersion Lithography



Immersion Lithography



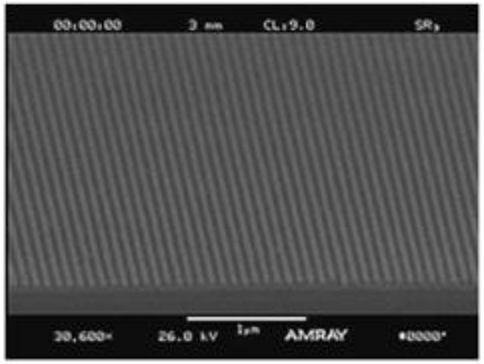
$NA = n\sin(\alpha)$



$$l_{min} = k_1\lambda/NA$$

For $k_1 = 0.25$
$\lambda = 193$ nm
$NA = 0.93$,

$n_{air} = 1$	$l_{min} = 51$ nm
$n_{water} = 1.44$	$l_{min} = 35$ nm



45 nm features

Enhancement Techniques

Immersion Lithography

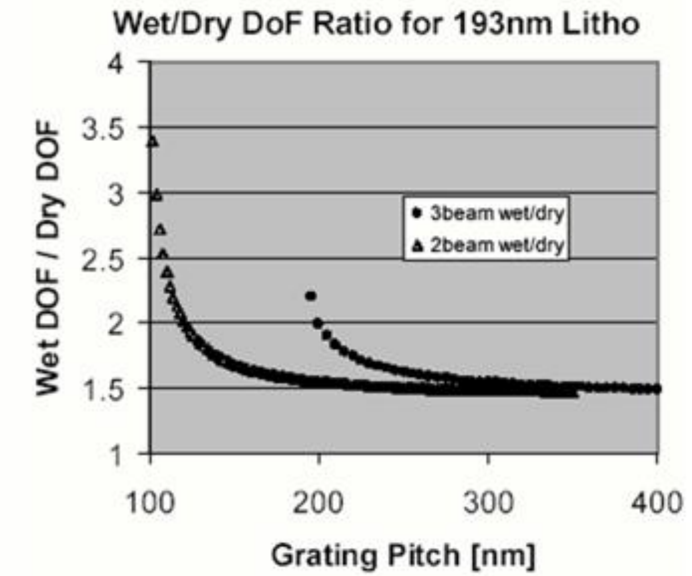
- Extend past NA > 1 limit
  - as high as n_{fluid} allows
- DOF improves by more than n
- H_2O is good choice at λ= 193 nm
  - $n_{water} \approx 1.44$

Challenges:

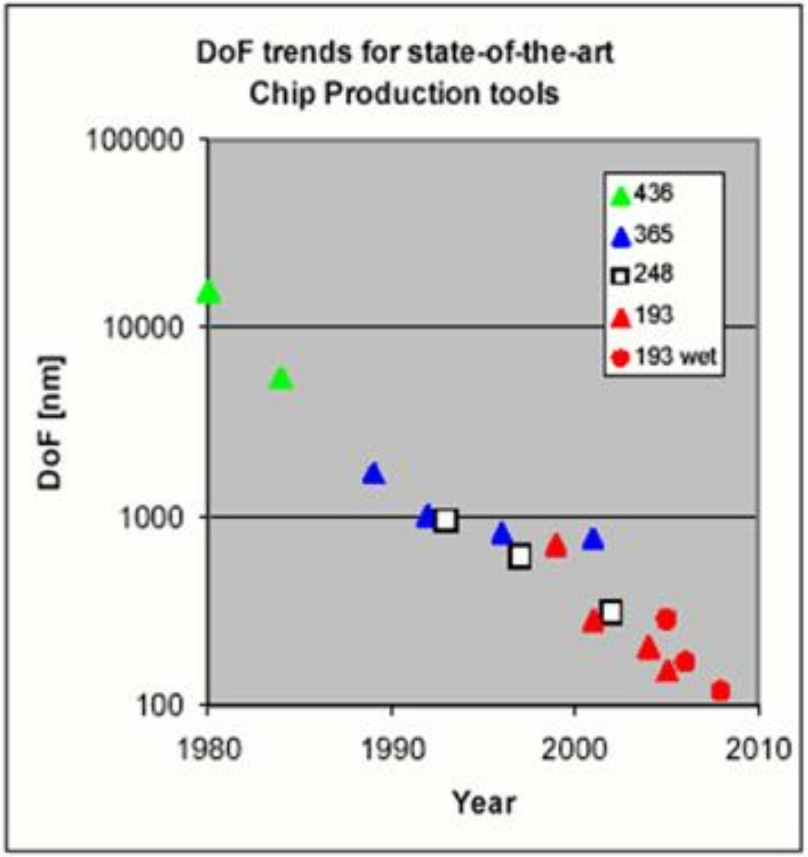
- Optics cost and complications
- Defects
- Contamination
- Fluid cost

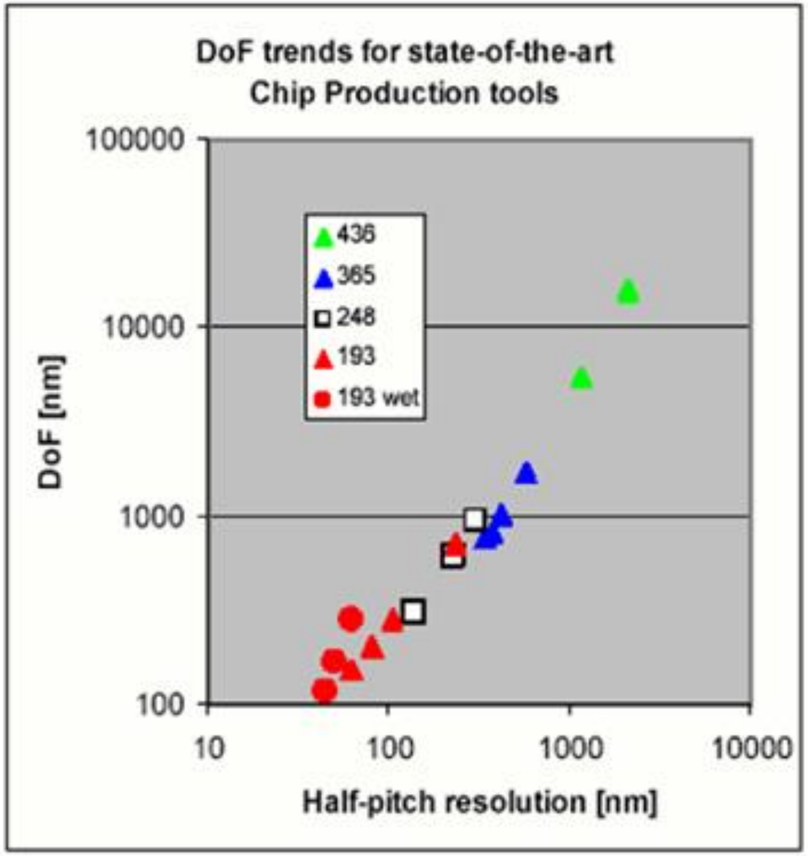
$$DOF = \frac{k_2}{2}\frac{\lambda}{n_{fluid}(1-\cos\theta)}$$

$$\frac{DOF(immersion)}{DOF(dry)} = \frac{1-\sqrt{1-(\lambda/p)^2}}{n_{fluid}-\sqrt{(n_{fluid})^2-(\lambda/p)^2}}$$



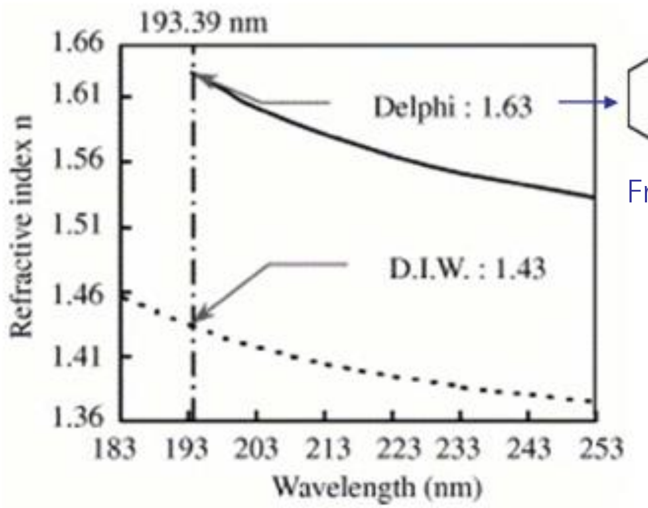
DOF Trends

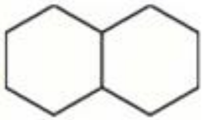

DoF trends for state-of-the-art
Chip Production tools
DoF [nm]
100000
10000
1000
100
1980
1990
2000
2010
Year
436
365
248
193
193 wet


DoF trends for state-of-the-art
Chip Production tools
DoF [nm]
100000
10000
1000
100
10
100
1000
10000
Half-pitch resolution [nm]
436
365
248
193
193 wet

Extending Immersion Lithography

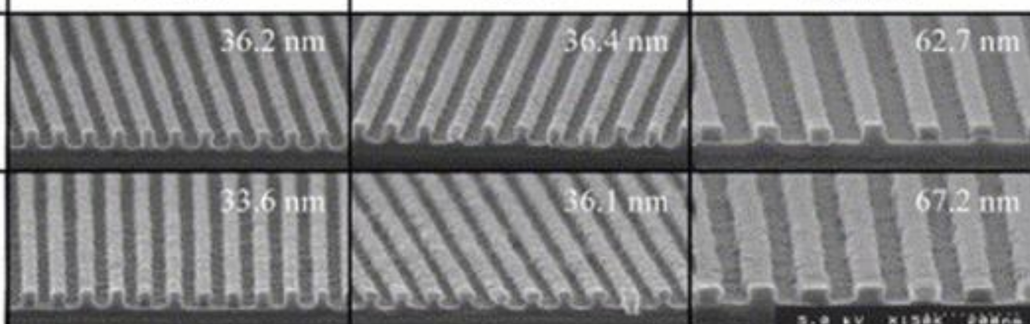
Higher index fluids





From Sumitomo

Topcoat / Photoresist	32 nm hp 'Delphi'	36 nm hp D.I.W.	60 nm Dry (Air)
TILC021 / ILP-07	36.2 nm	36.4 nm	62.7 nm
TILC021 / TArF-P6127	33.6 nm	36.1 nm	67.2 nm

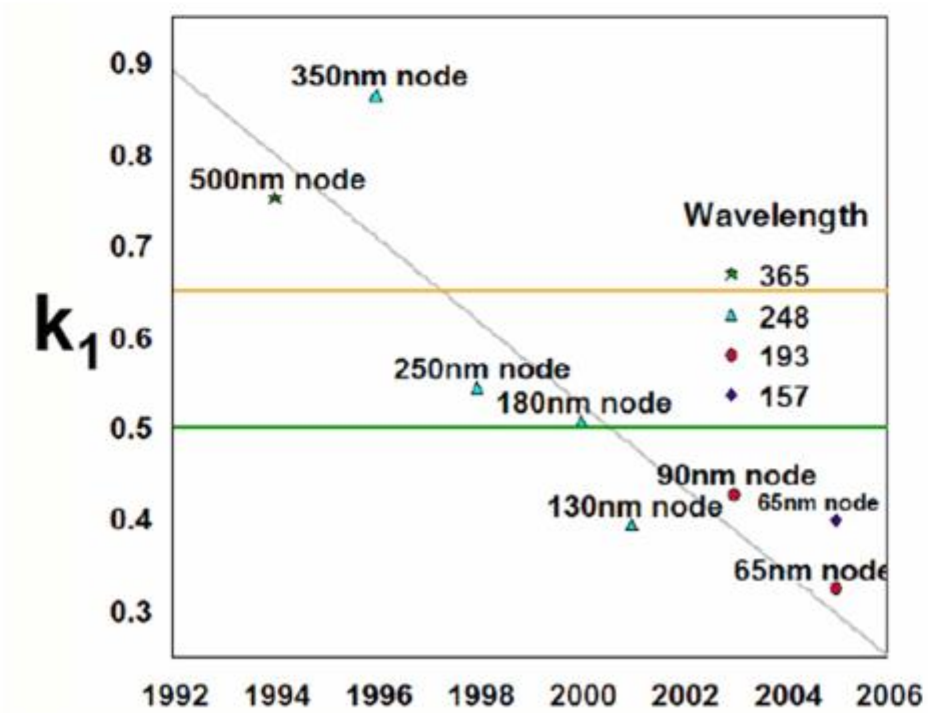


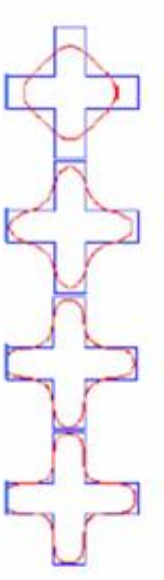
IMEC

Enhancement Techniques

Reducing k1

$l_{min} = k_1 \lambda / NA$





Keys:

- Low aberration optics
- Excellent process control
- Accurate mask patterns

Enhancement Techniques

Lithography Trends: k1, λ, NA

Year	λ	NA	fluid n	k1		DoF
1980	436	0.166	1	0.8	2101	15713
1984	436	0.28	1	0.75	1168	5450
1988	254	0.166	1	0.6	918	9154
1989	365	0.45	1	0.7	568	1706
1992	365	0.57	1	0.65	416	1023
1993	254	0.5	1	0.6	305	948
1996	365	0.63	1	0.65	377	817
1997	248	0.6	1	0.55	227	620
1999	193	0.5	1	0.6	232	720
2001	193	0.75	1	0.4	103	285
2001	365	0.65	1	0.6	337	760
2002	248	0.8	1	0.45	140	310
2004	193	0.85	1	0.35	79	204
2005	193	0.93	1	0.3	62	153
2005	193	0.93	1.44	0.3	62	283
2006	193	1.15	1.44	0.3	50	168
2008	193	1.3	1.44	0.3	45	118

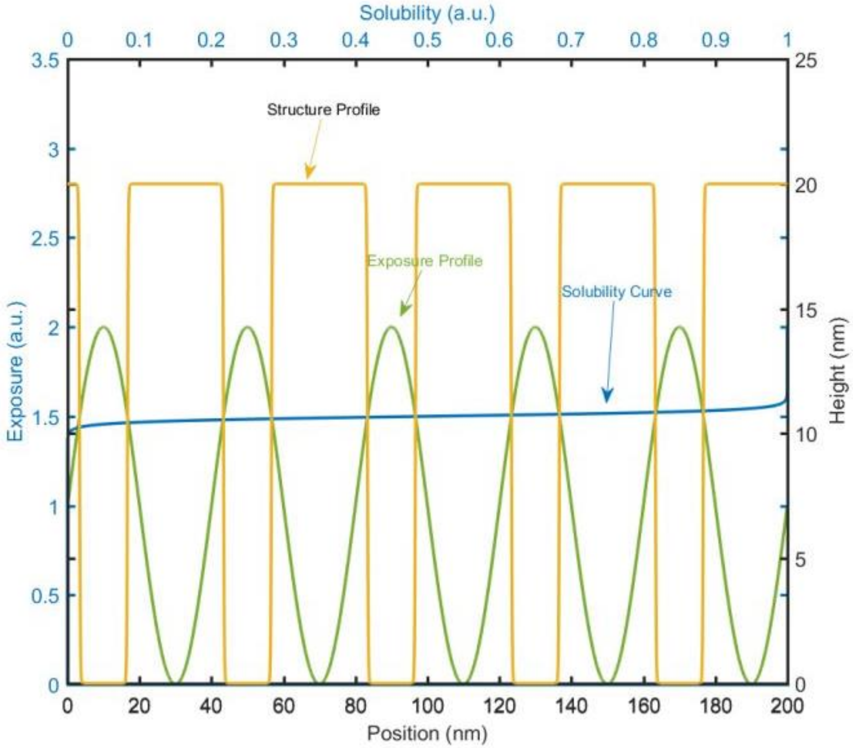
All dimensions in nm

Resolution Enhancement Technologies

(RET)

Improved resolution and DOF

- PSM –Phase shift mask
  - –Attenuated
  - – Alternating
- OAI – Off axis illumination
- SRAF –
  Sub-resolution assist features



八仙过海各显神通

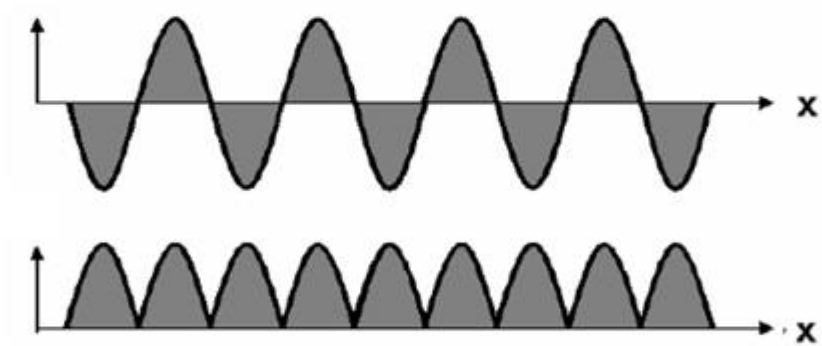
Phase Shifting Masks

Optical intensity and exposure energy

Electric field (ε): $$\varepsilon(\vec{r},\nu)=\varepsilon_0(\vec{r})e^{j\phi(\vec{r},\nu)}$$

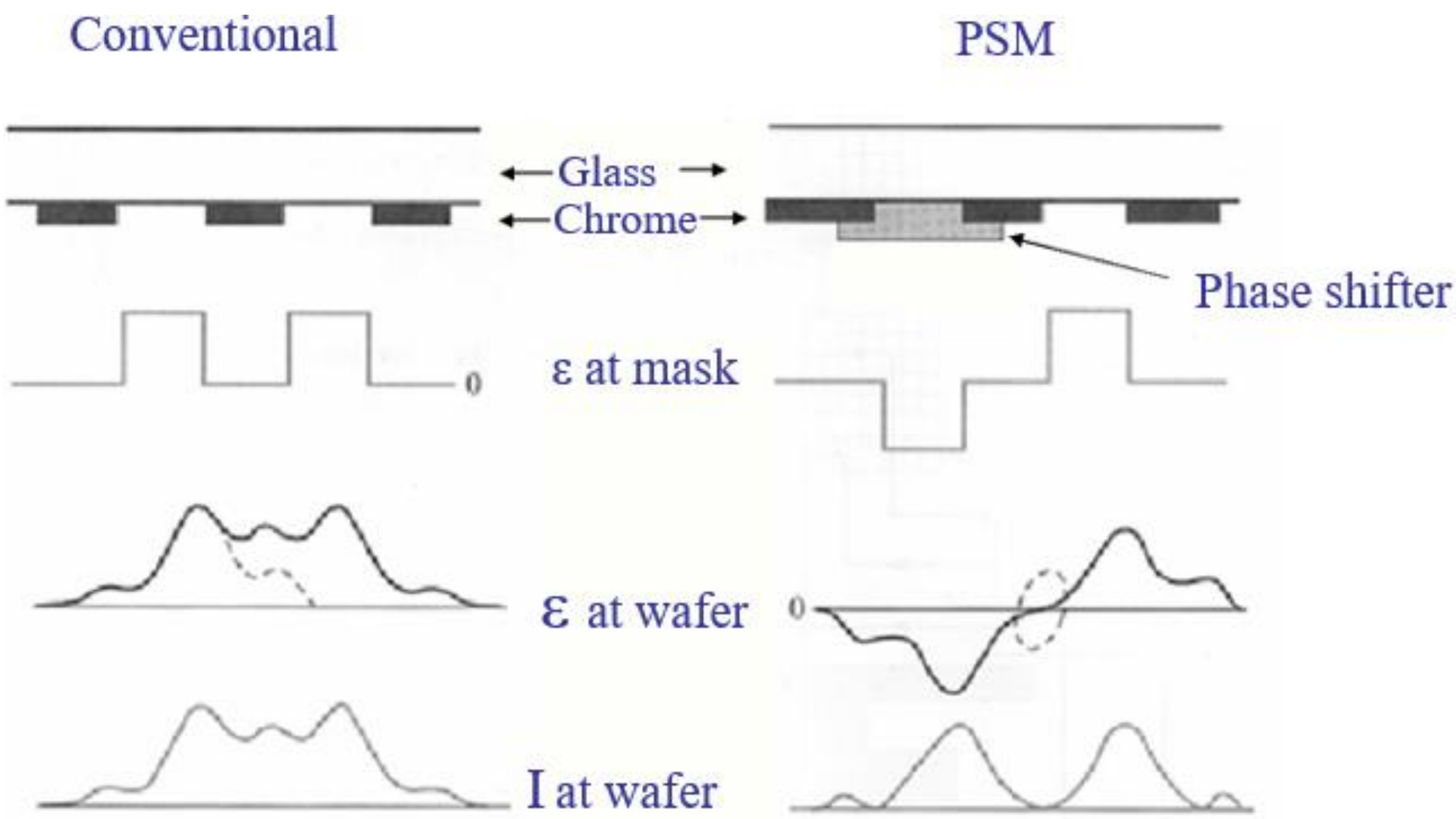
Intensity (I): $$I=\varepsilon\cdot\varepsilon^{*}=\phi_0 e^{j\phi}\cdot\varepsilon_0 e^{-j\phi}=\varepsilon_0^2$$

Energy (E): $$E=\int I\cdot dt\approx I\cdot t$$

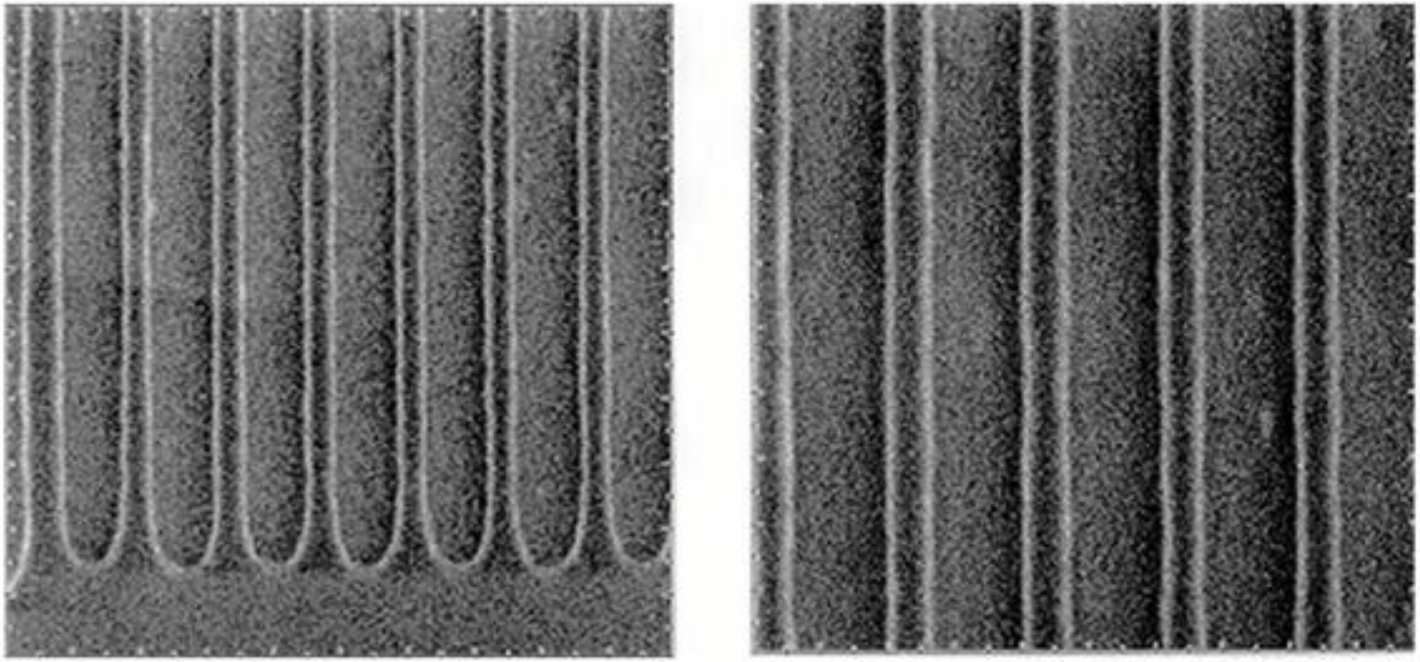


通过一透明区的光波与通过相邻透明区但具有**180°相位差**的光波之间的干涉作用,使**相邻透明区边界处的光强减弱**,从而使图形边沿**对比度**得到提高.

Phase Shifting Lithography



Phase Shifting Lithography

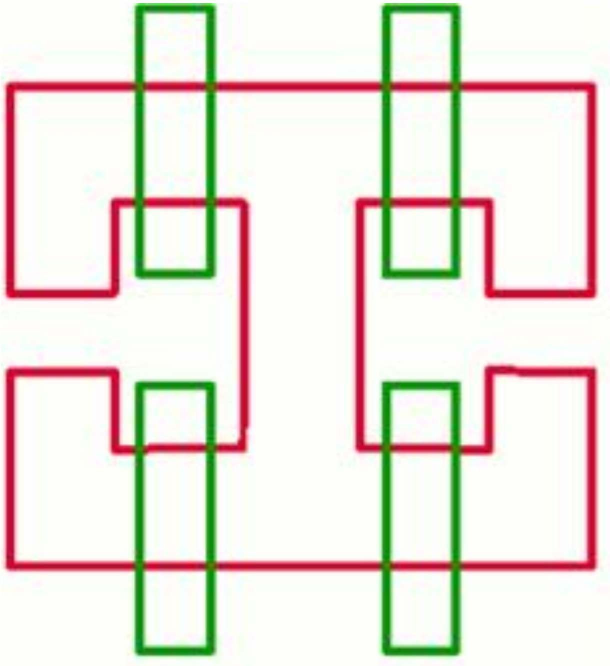


60 nm lines printed with 193 nm DUV lithography and PSM

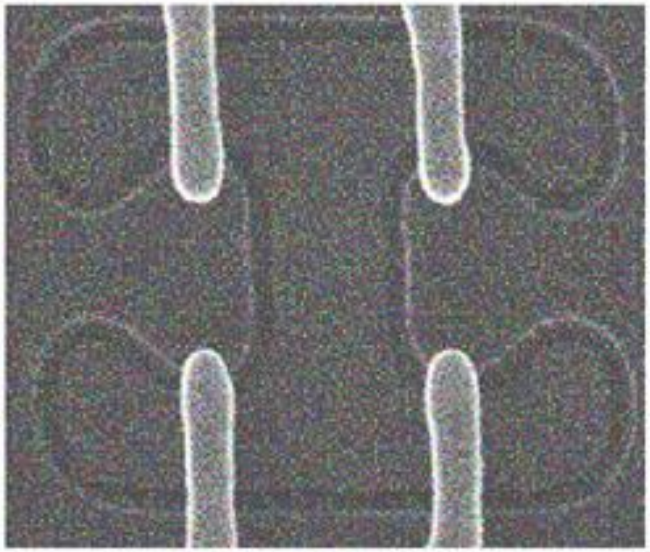
Lucent

Optical Proximity Correction (OPC)

CAD

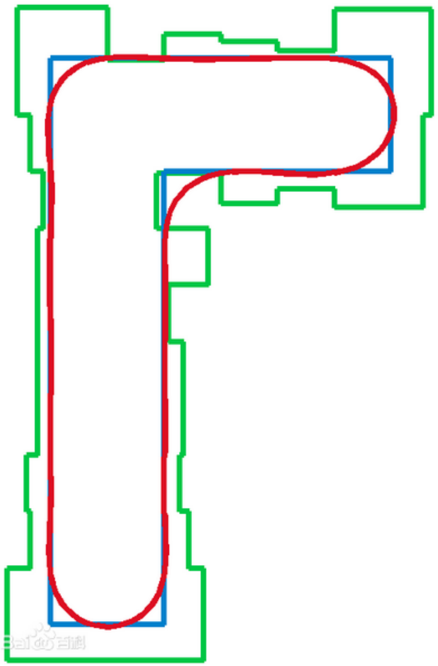


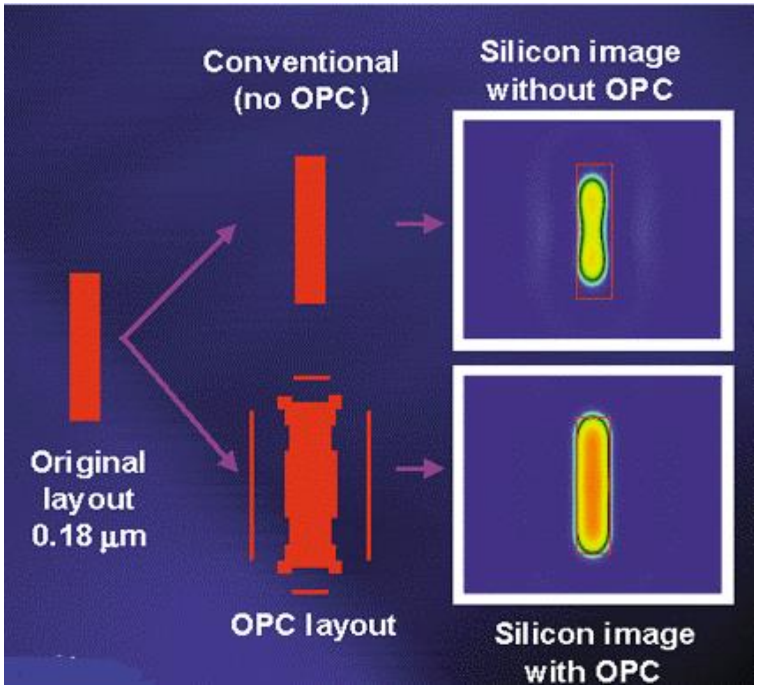
After process

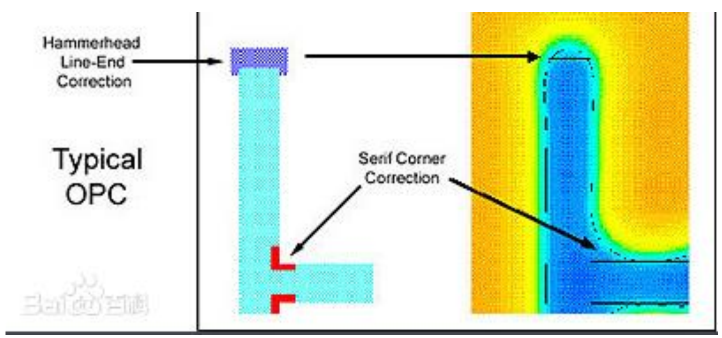


- Corner rounding
- Line shortening
- Overlay errors
- CD errors
- Bends and wiggles

Optical Proximity Correction (OPC)



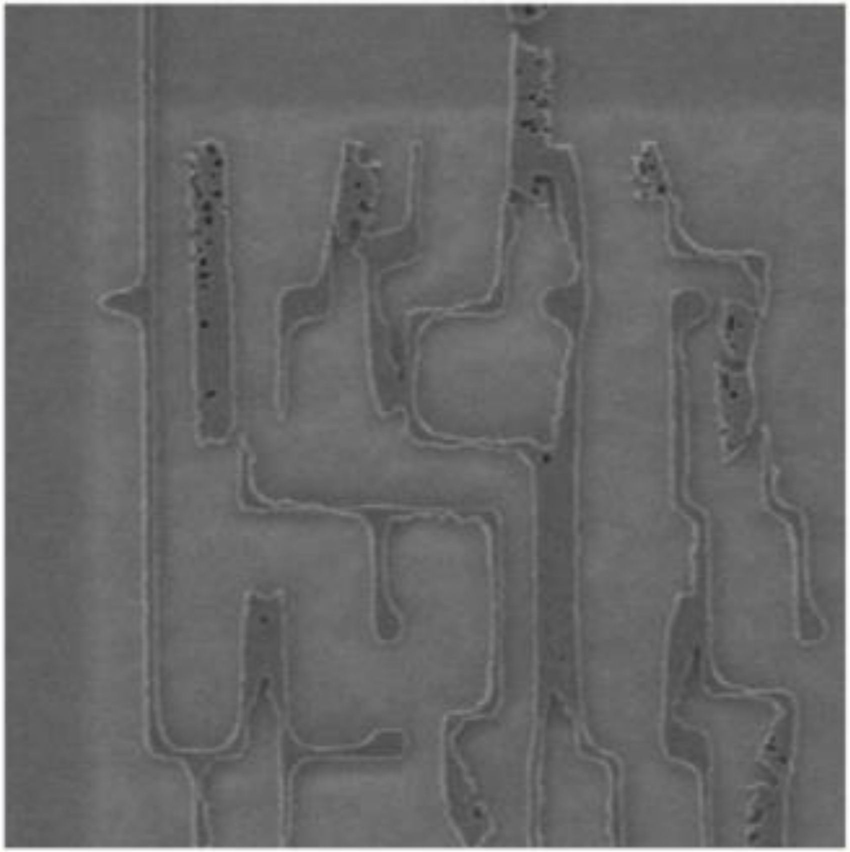




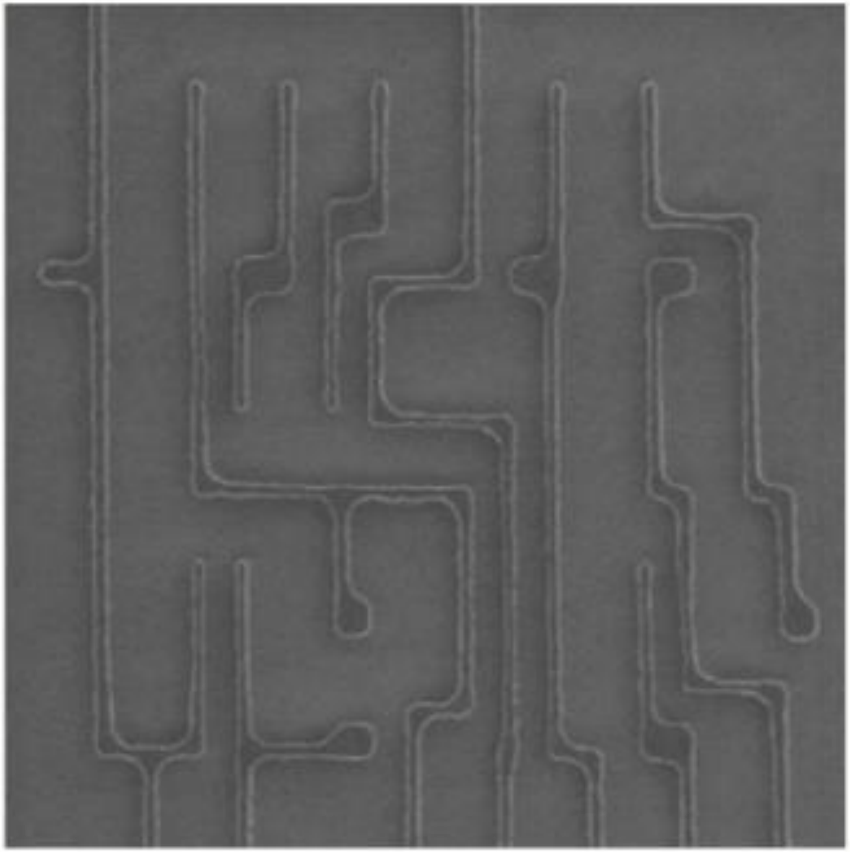
在光刻版上进行图形修正，补偿衍射带来的光刻图形变形。

Optical Proximity Correction (OPC)

No OPC



OPC

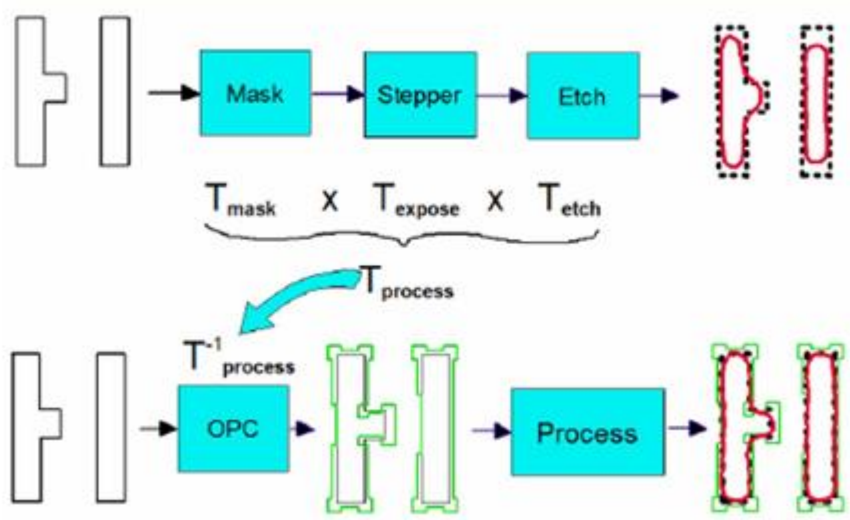


OPC: Total Process Compensation

Intentional ***distortion of mask pattern*** in order to compensate for systematic and stable patterning errors.

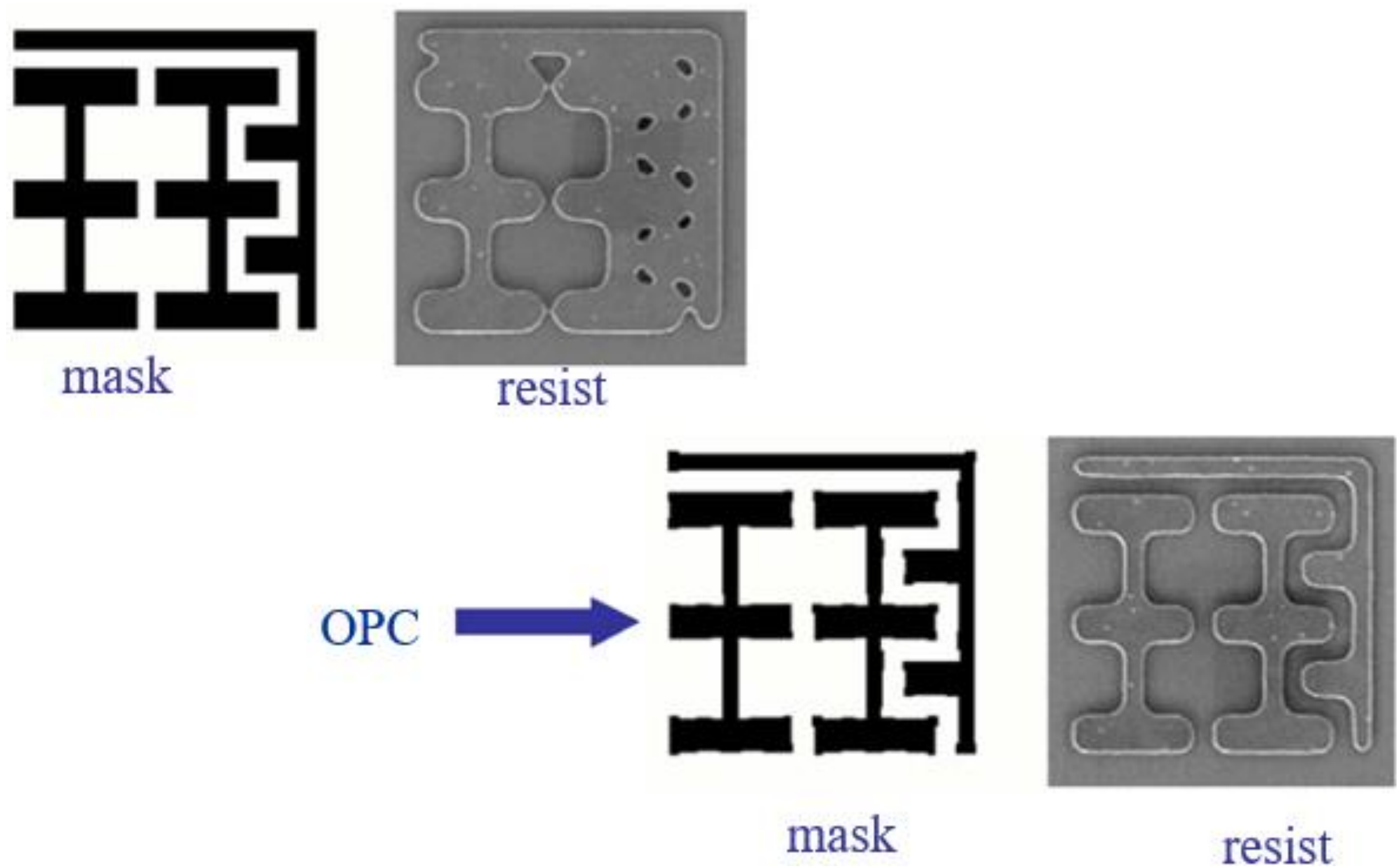
Historical Progression:

- "Tweaking" of DRAM and SRAM cells
- Anchor shapes to compensate for gate shortening
- Rules-based OPC
  - selective line bias according to correction table
  - implemented with design rule checker
- Model-based OPC
  - iterative shape modification until model fits tolerances



Enhancement Techniques

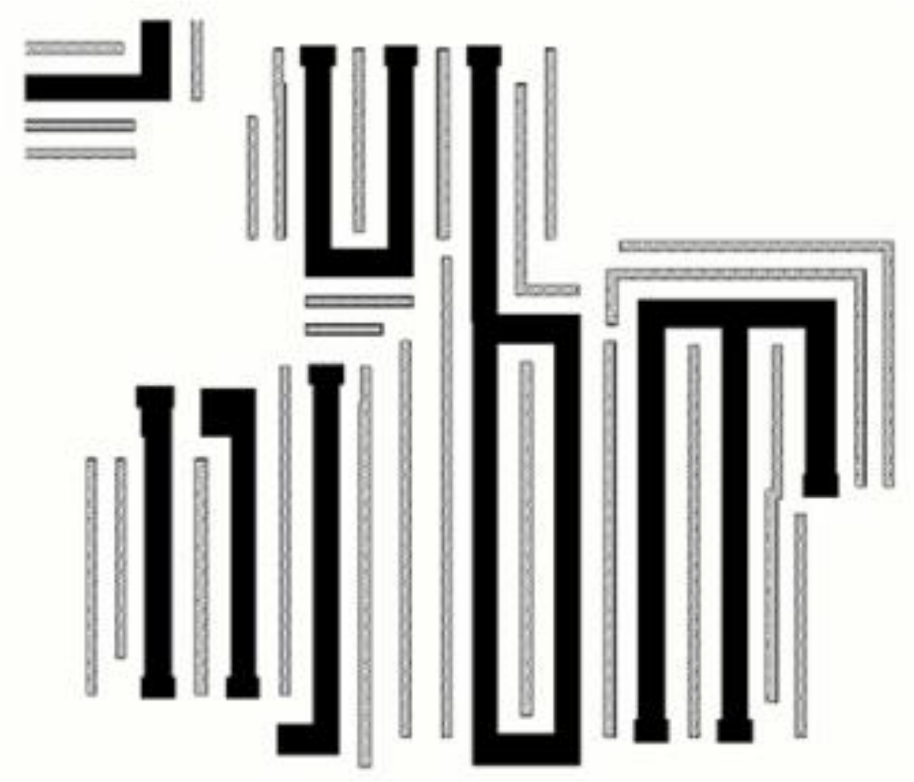
Optical Proximity Correction



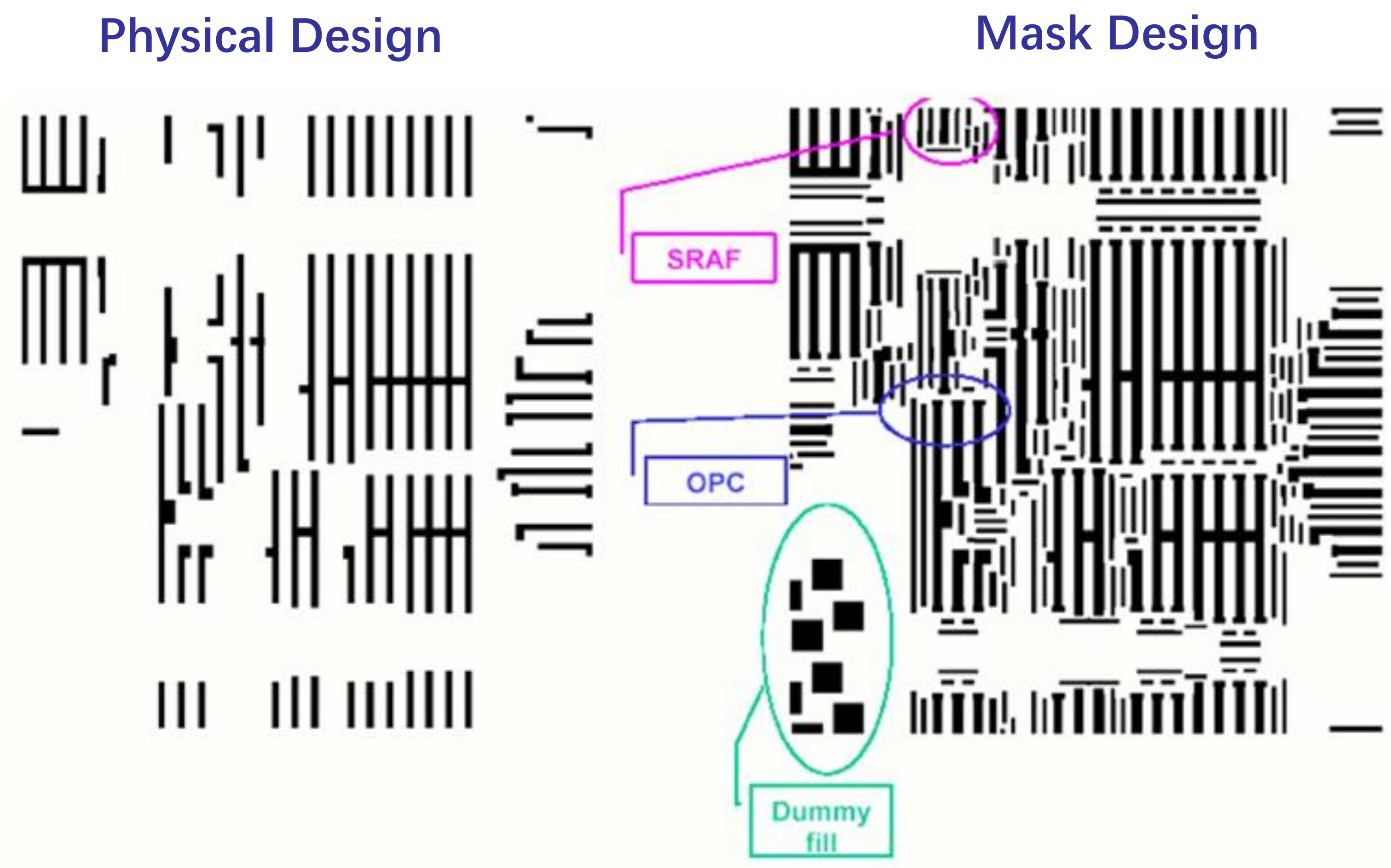
Sub-Resolution Assist Features

(SRAFs)

- Main features print with larger DOF
- Assist features should not print
- Isolated lines and dense lines become more uniform



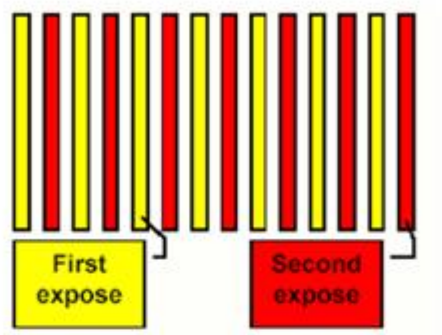
Data Preparation

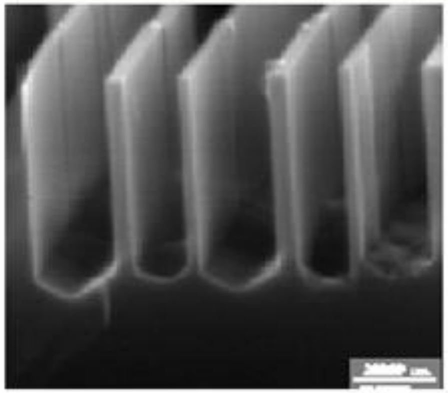


Double Patterning

- Double expose + double etch process
  - 2 gratings (1:3) interleaved

  (will not work with simple double exposu
- Reduces effective k1:
  - Practical printing of half-pitch $k_1 = 0.15$
  - 2 $k_1 = 0.3$ gratings
- Subject to overlay errors and processing distortions

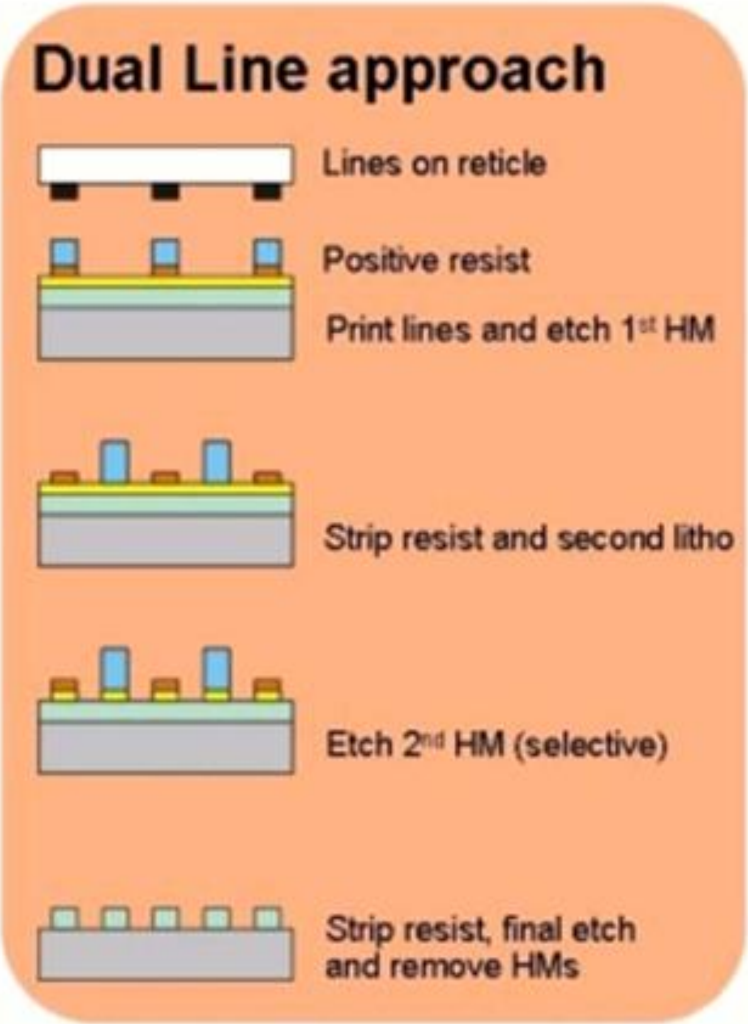




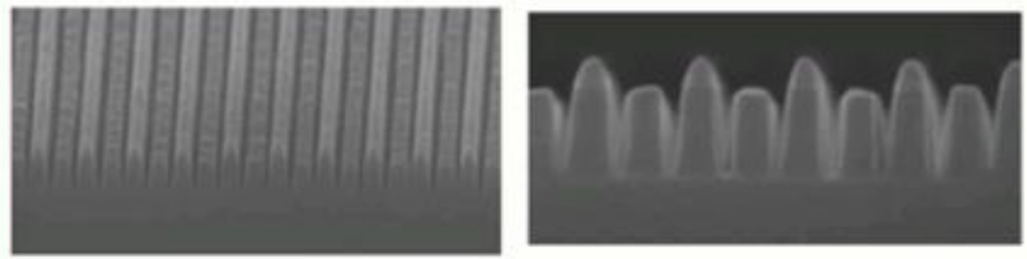
180 nm pitch grating from two exposures of 360 nm grating.

(Brueck et al., SPIE 2002)

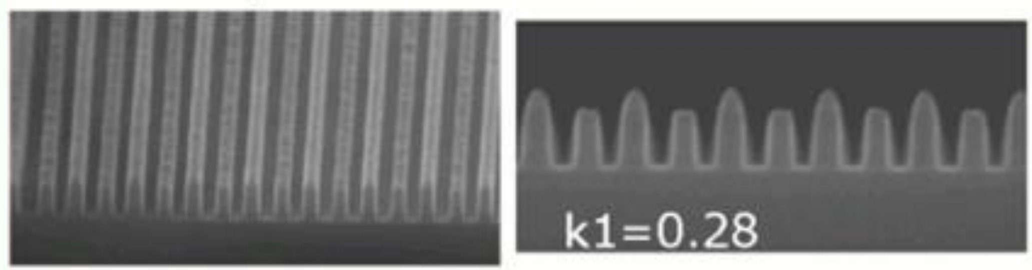
Resist Trim



- 32nm hp → FLASH 32nm node

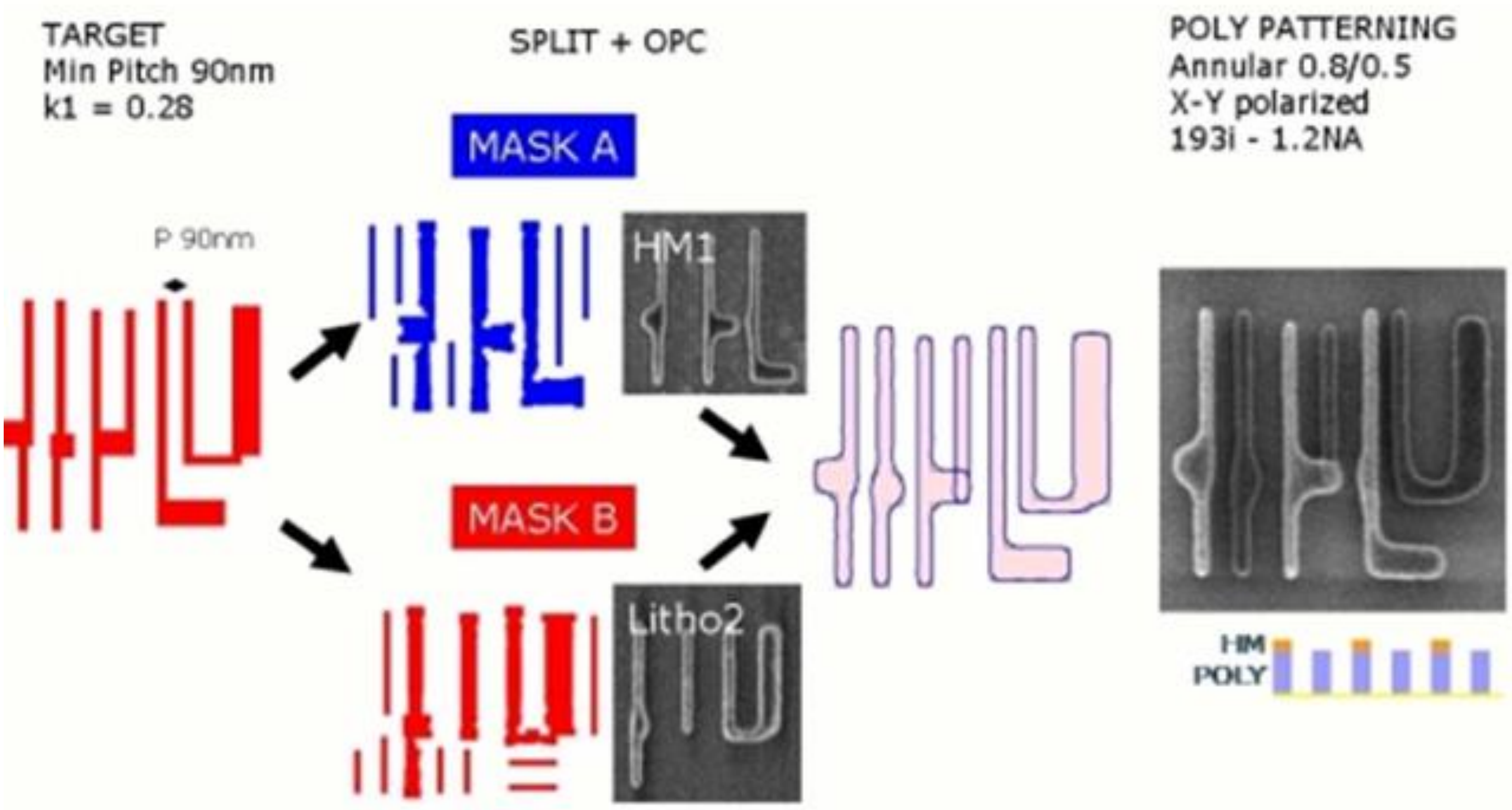


- 44nm hp → LOGIC 32nm node



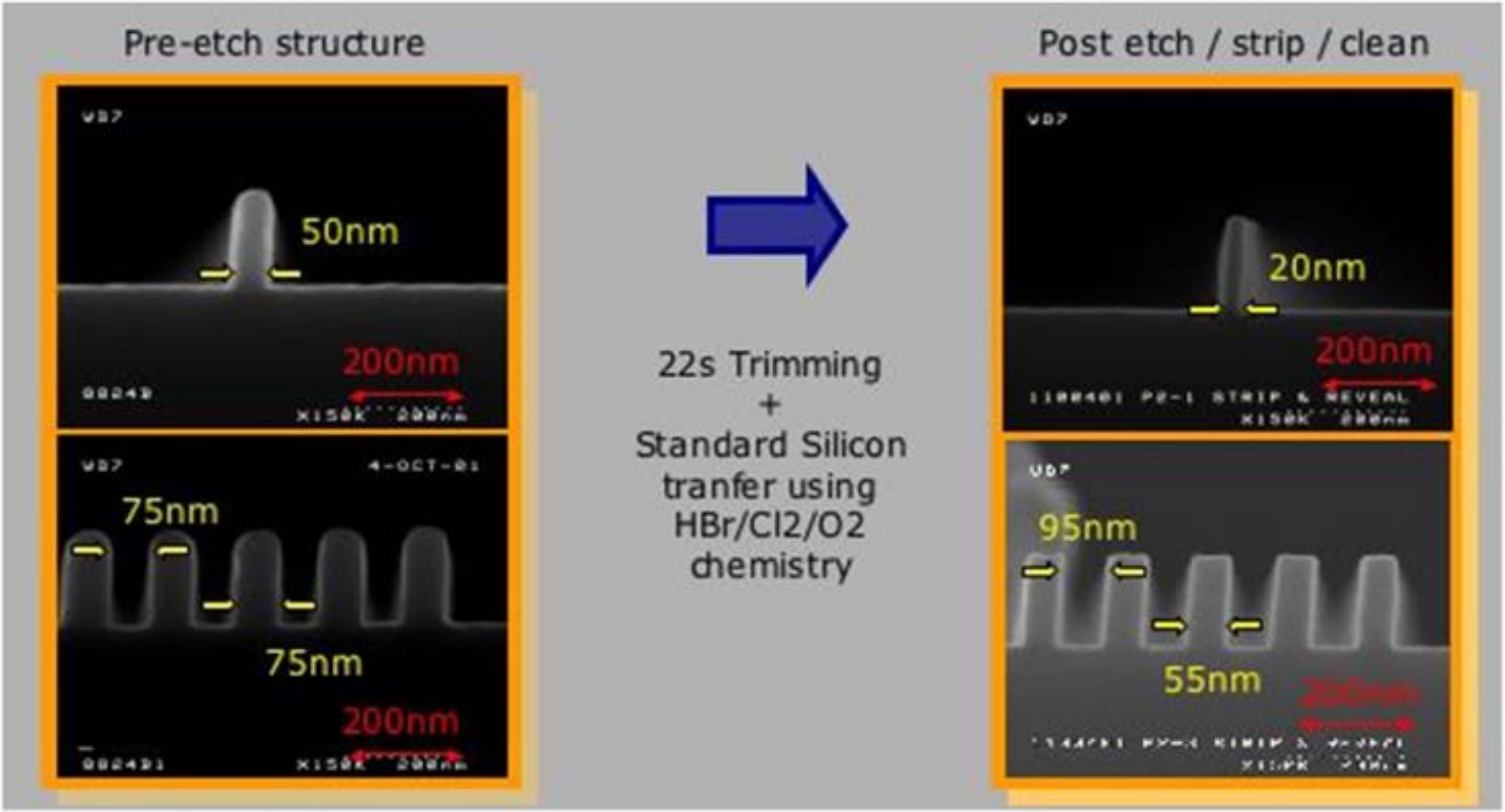
IMEC

Double Patterning + OPC



IMEC

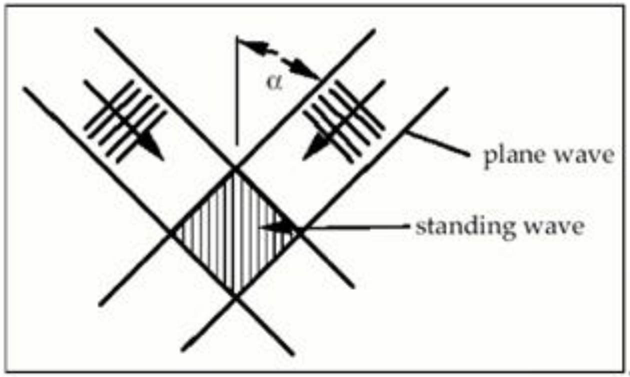
Resist Trim



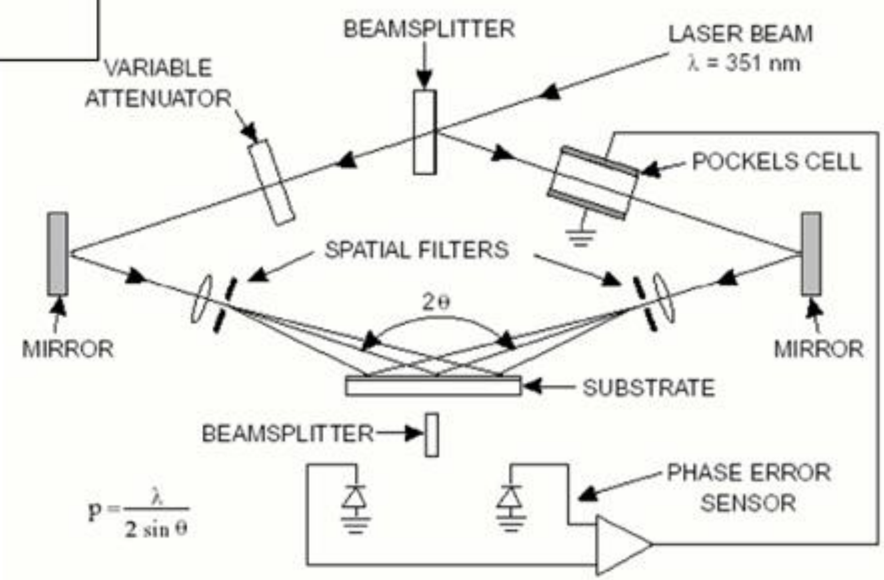
灰化工艺、侧壁工艺降低k1

Enhancement Techniques

Interferometric Lithography



$P_L = \lambda/2 \sin\alpha$

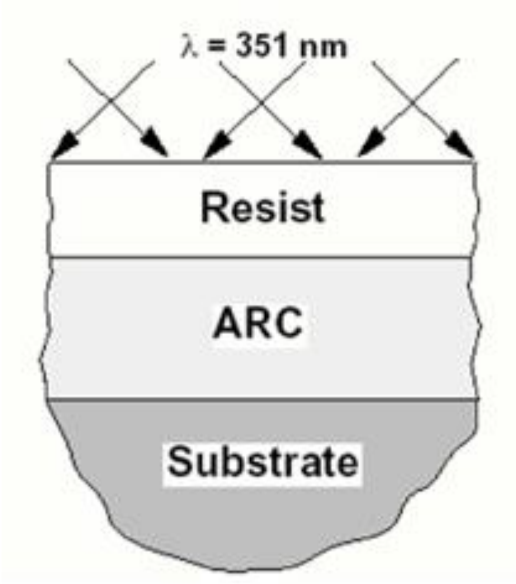


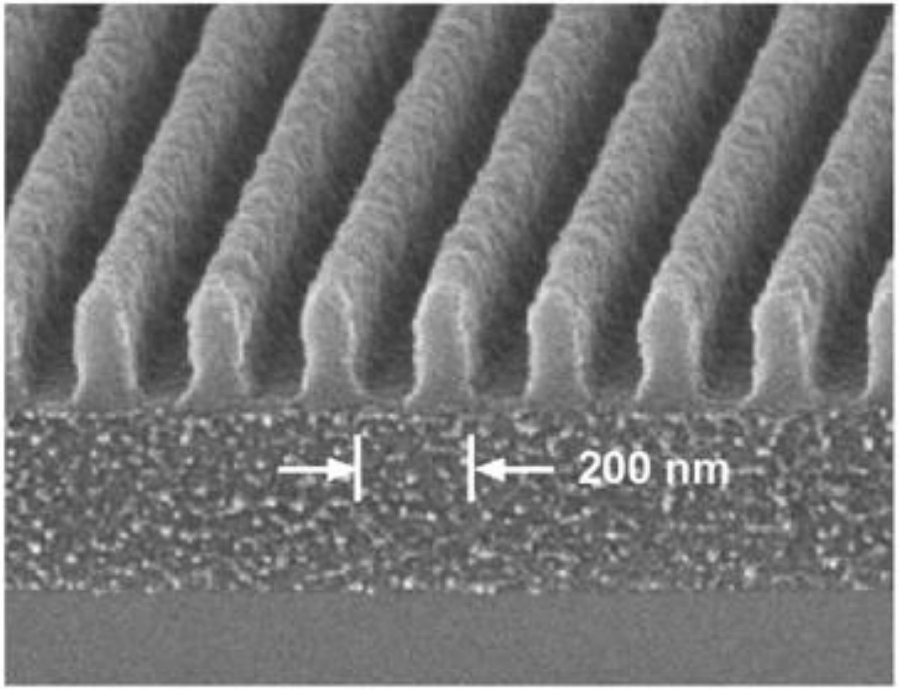
- Maskless
- Depends only on λ and α
- Only periodic structures (no arbitrary features)

PSM、双重掩膜、OPC、驻波效应校正属于波前工程突破

Enhancement Techniques

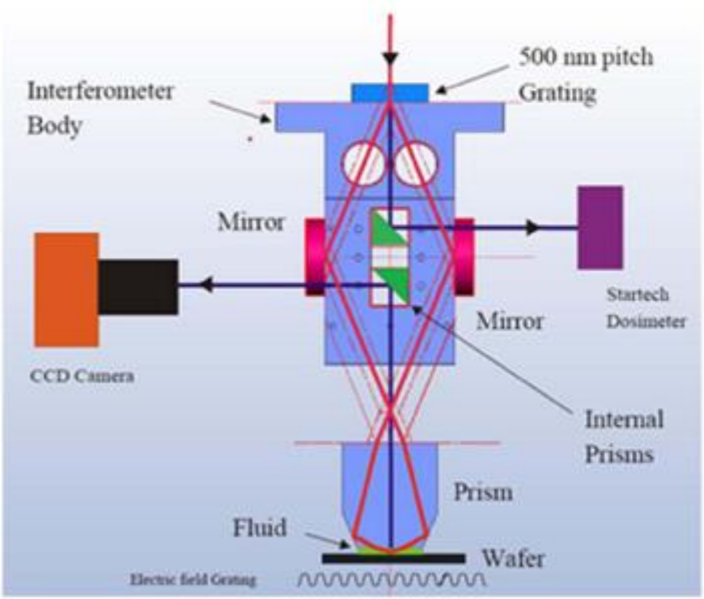
Interferometric Lithography

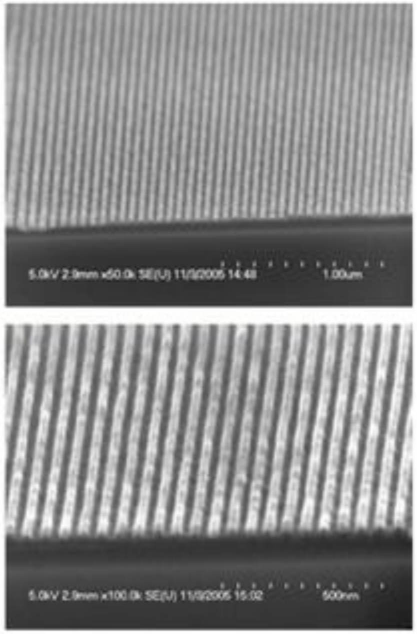




Immersion + Interference Lithography

NEMO System -
IBM Almaden

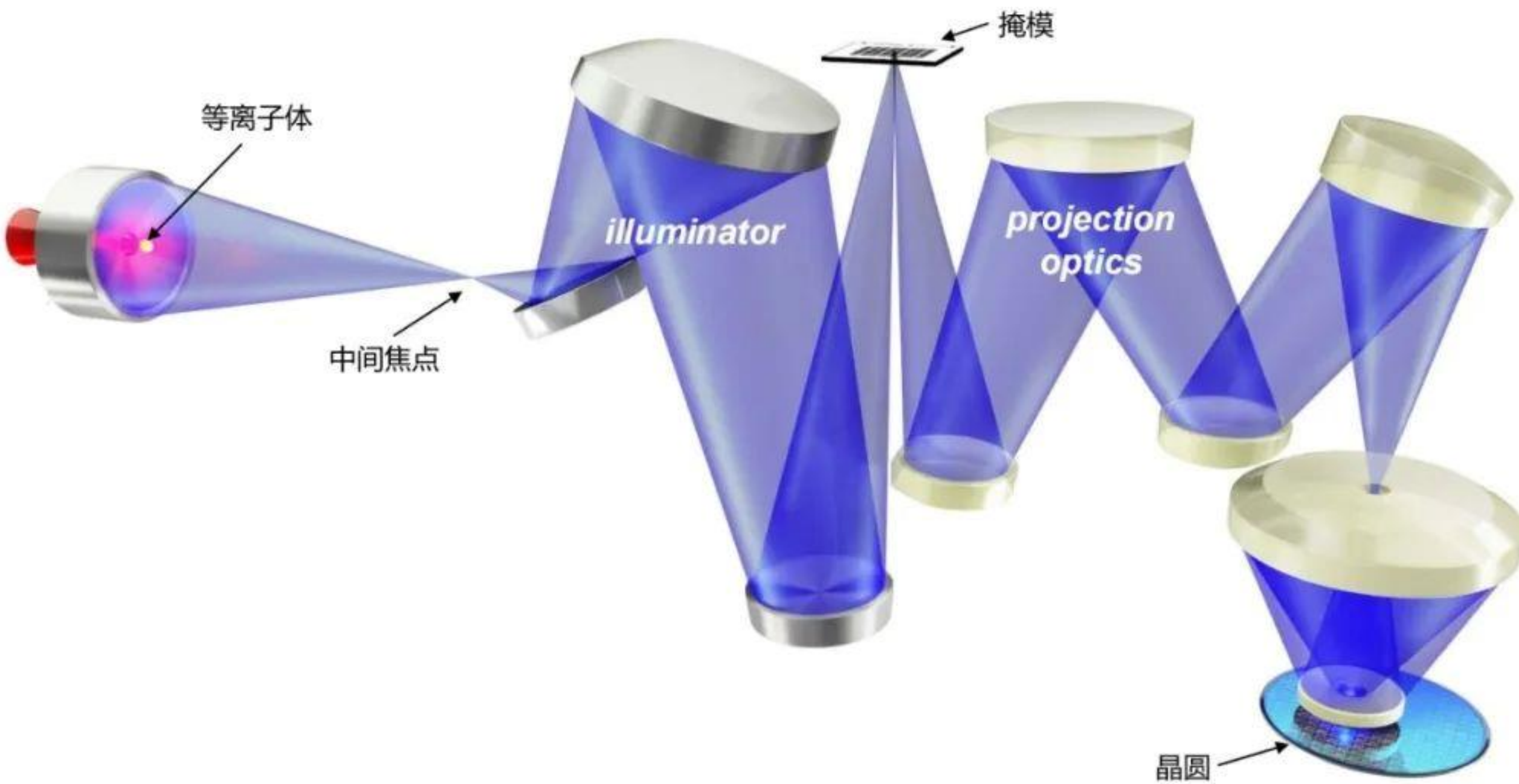


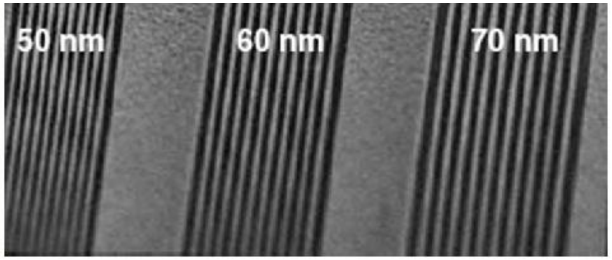


30 nm half pitch

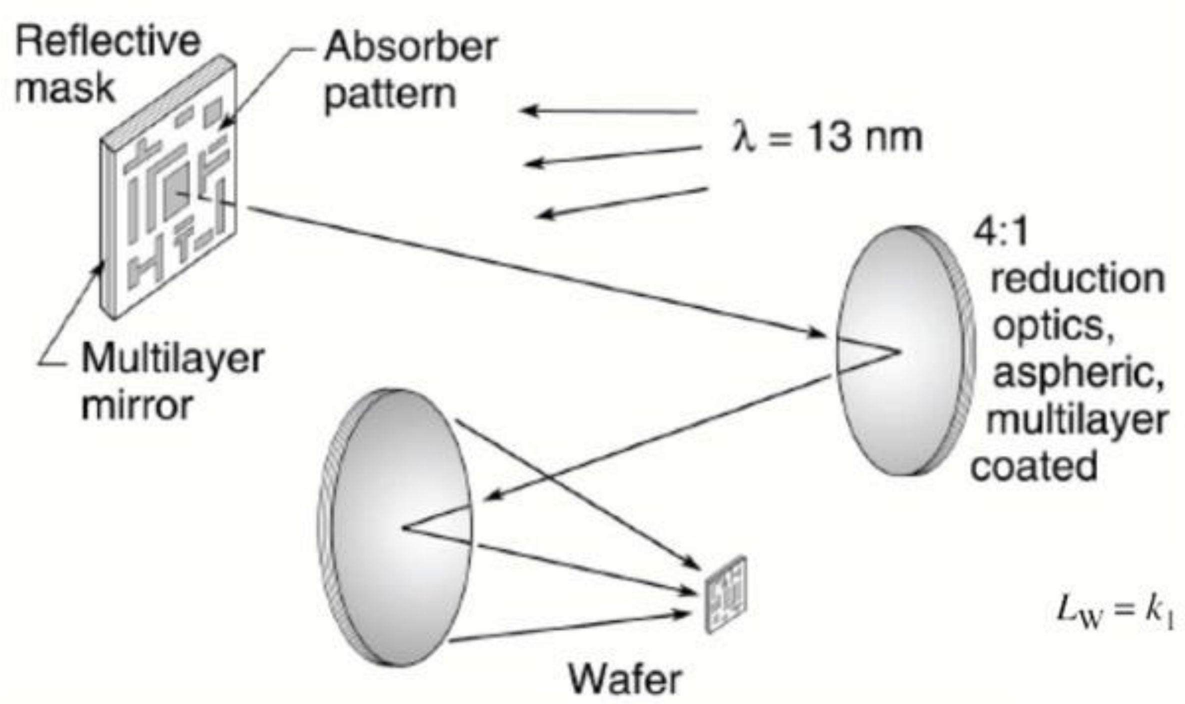
Outline

EUV


掩模
等离子体
illuminator
projection optics
中间焦点
晶圆


50 nm
60 nm
70 nm

EUV Lithography

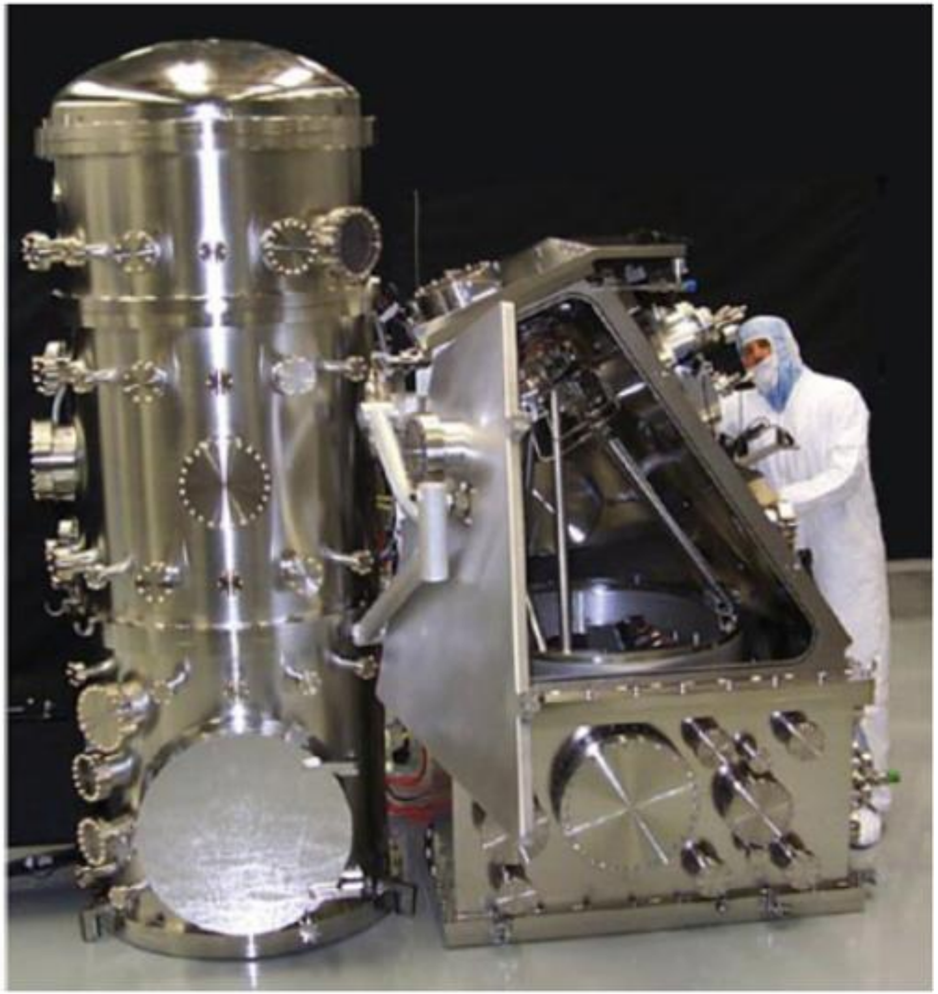


$$L_W = k_1 \frac{\lambda}{NA}$$

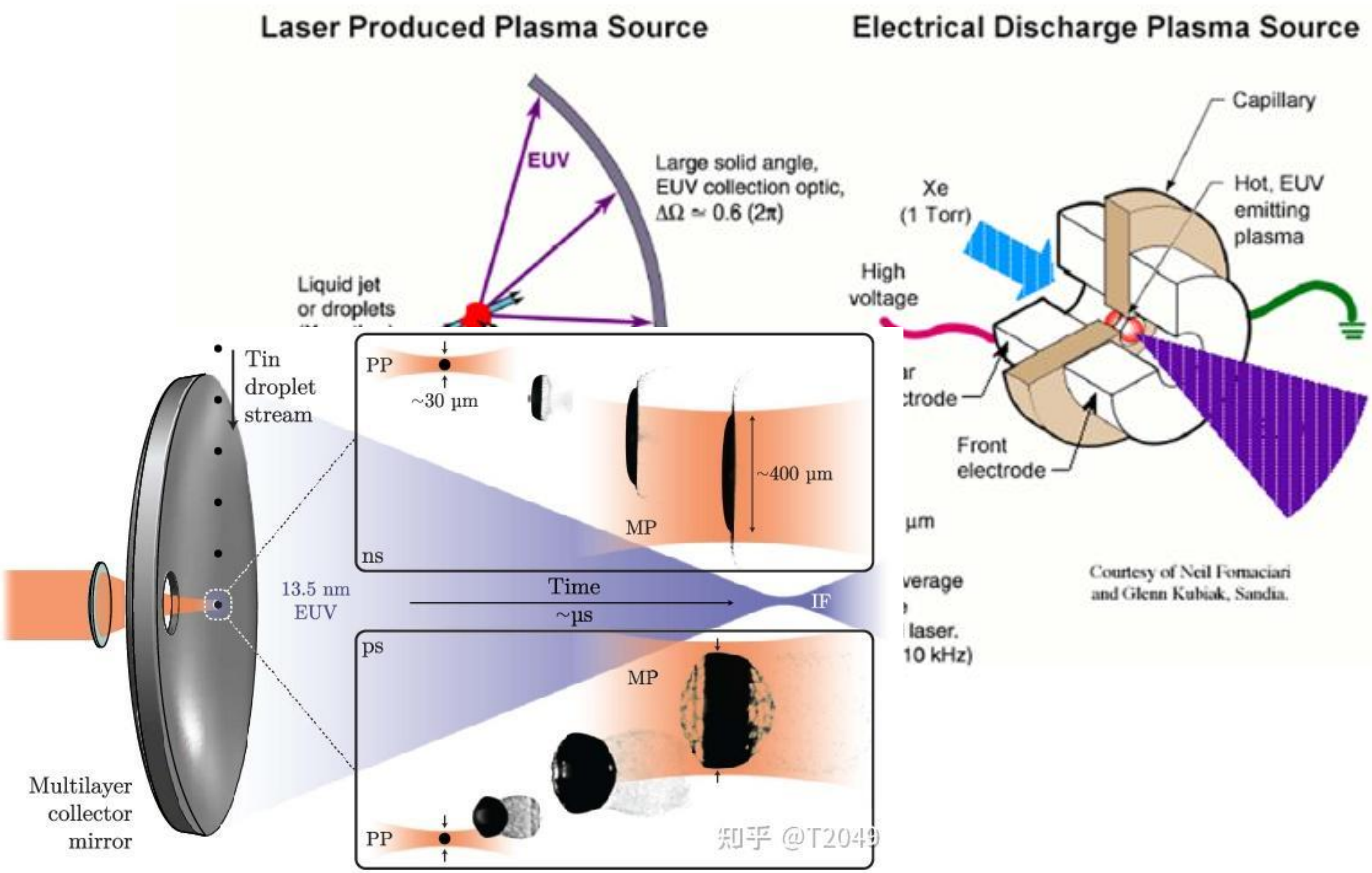
$$DOF = \pm k_2 \frac{\lambda}{(NA)^2}$$

$$\sigma = \frac{NA_{cond}}{NA_{obj}}$$

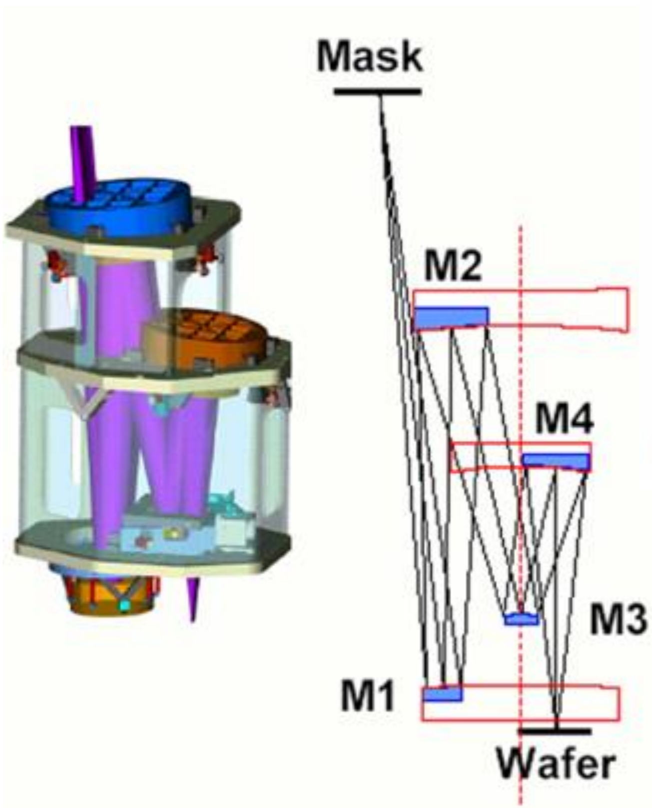
EUV Engineering Test Stand
(LLNL)

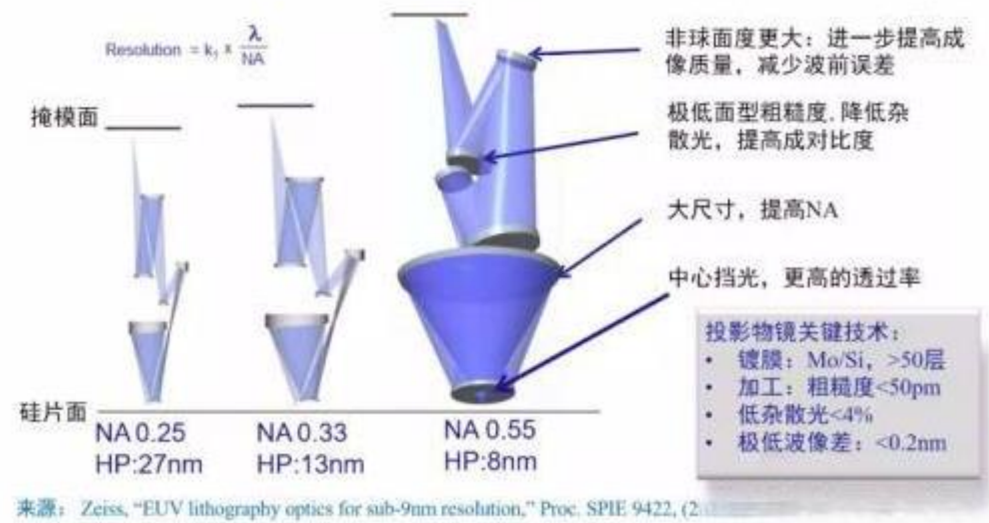


EUV Sources


Laser Produced Plasma Source
Electrical Discharge Plasma Source
EUV
Large solid angle,
EUV collection optic,
ΔΩ ≈ 0.6 (2π)
Liquid jet
or droplets
Capillary
Hot, EUV
emitting
plasma
Xe
(1 Torr)
High
voltage
Front
electrode
Courtesy of Neil Fornaciari
and Glenn Kubiak, Sandia.
Tin
droplet
stream
PP
~30 µm
~400 µm
MP
ns
Time
~µs
IF
ps
MP
PP
13.5 nm
EUV
Multilayer
collector
mirror
知乎 @T2049

ETS Optics





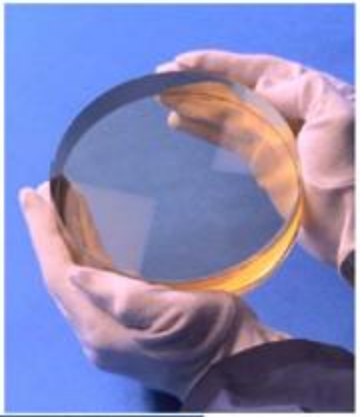
0.1 N.A.
λ/20 RMS wavefront error
24 mm x 1.5 mm scanning ring field
< 1 nm dynamic distortion

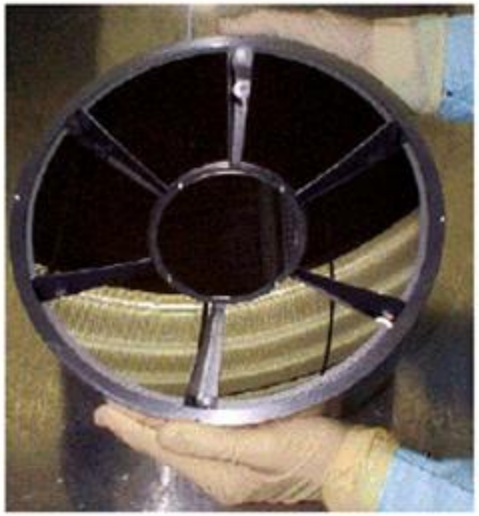
~1 meter overall length

EUV Optical Elements

Optics must be close to perfect

- Imperfections ~ $\lambda/50$
- Ultra smooth
- Low flare

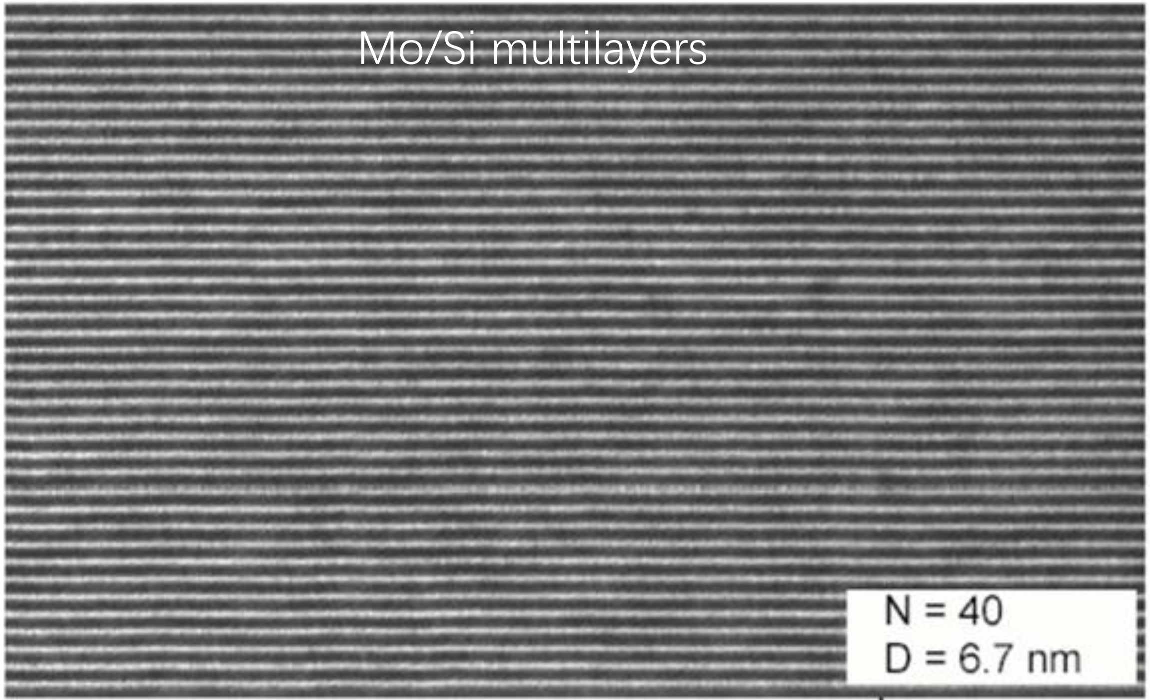




Condenser optic

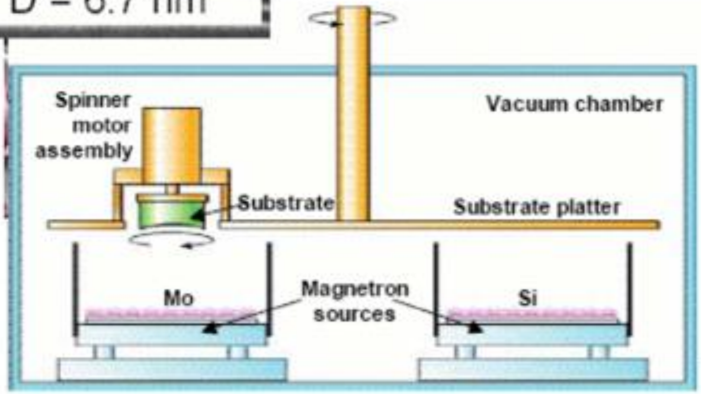
Projection optic

High Quality Multilayer Mirror

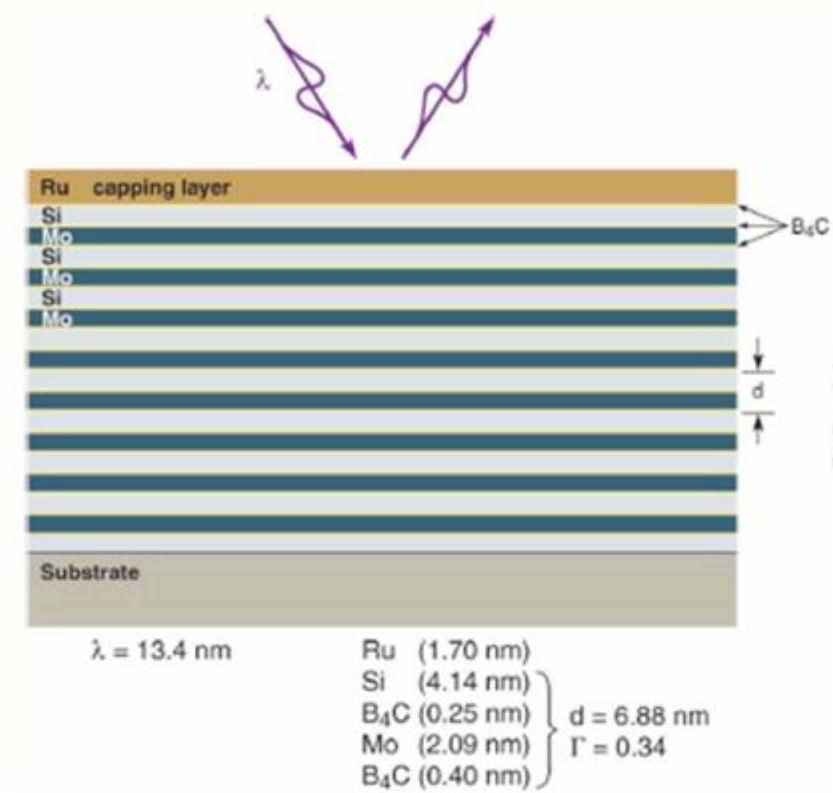


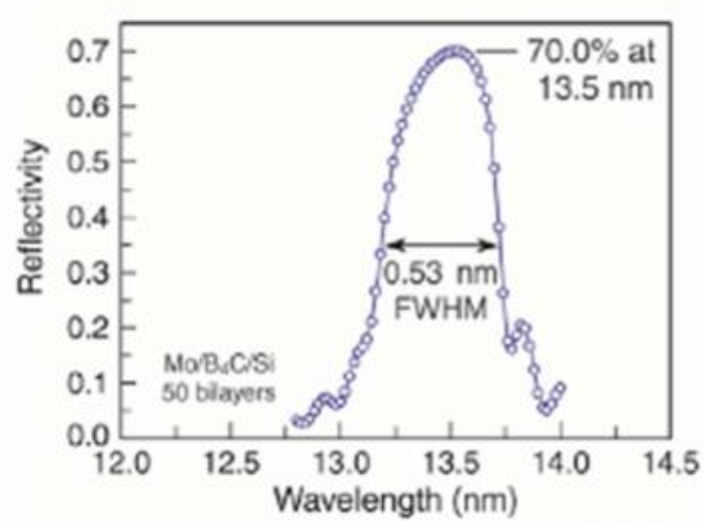
Multilayer coatings must be sufficiently uniform that they do not contribute to top surface error budget.

→ $\triangle t < \lambda/500$

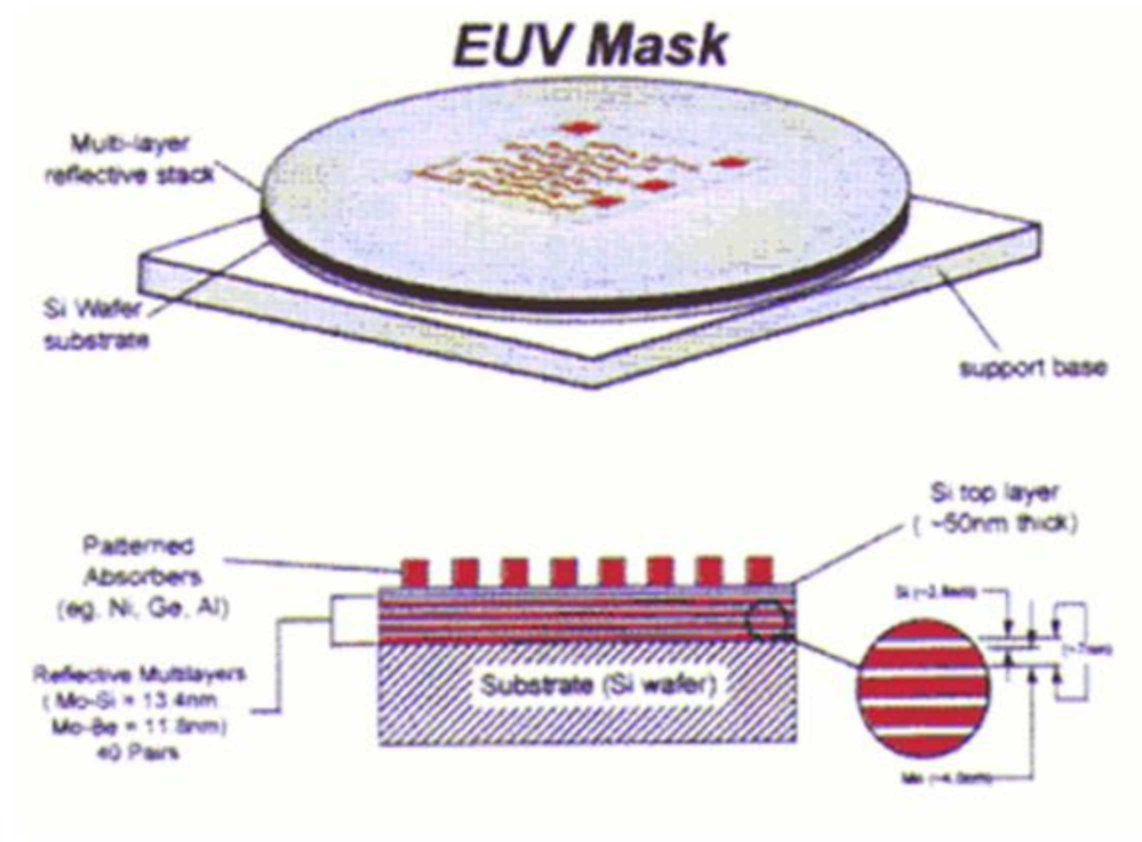


EUV Multilayer Optics

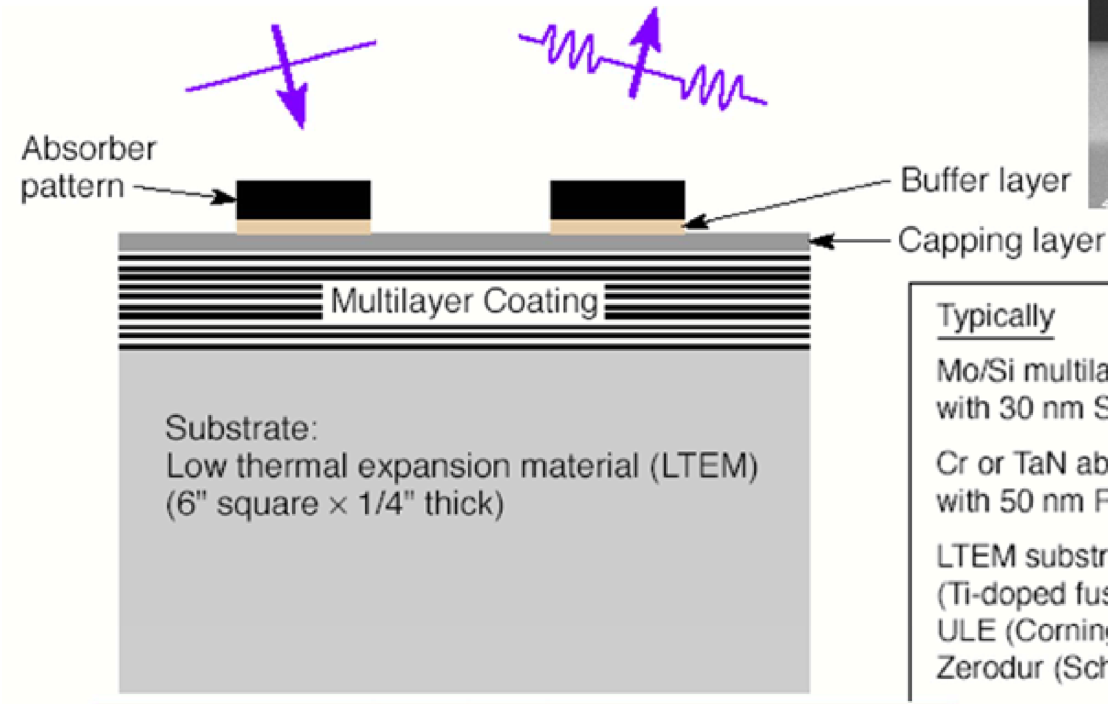


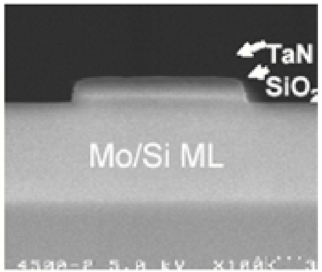


EUV Mask


EUV Mask
Multi-layer reflective stack
Si Wafer substrate
support base
Patterned Absorbers (eg. Ni, Ge, Al)
Reflective Multilayers (Mo-Si = 13.4nm Mo-Be = 11.8nm) 40 Pairs
Substrate (Si wafer)
Si top layer (~50nm thick)

EUV Mask





Typically

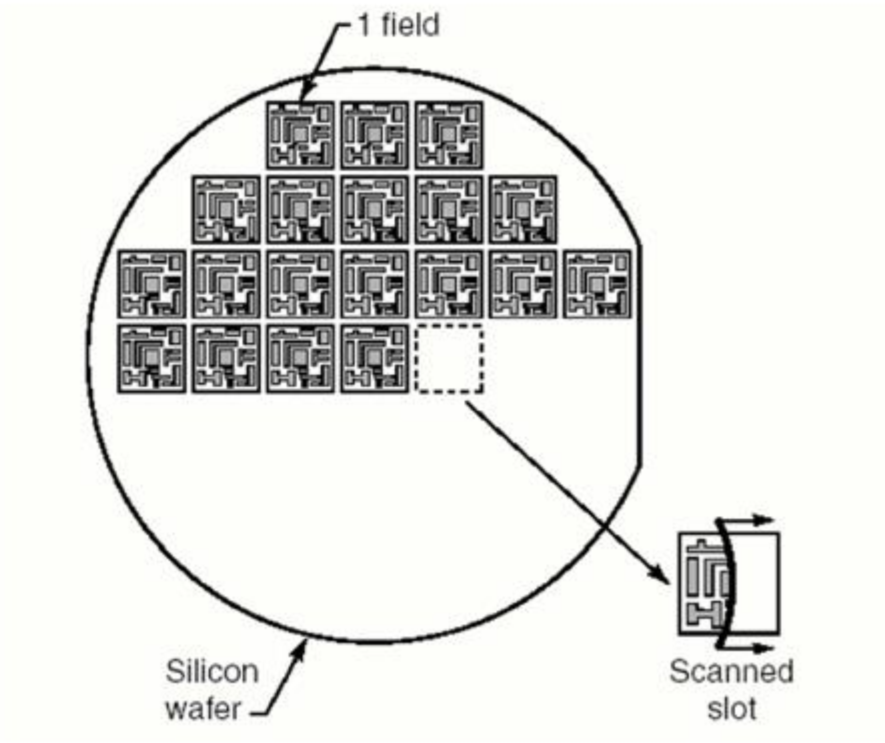
Mo/Si multilayer (d = 6.7 nm) with 30 nm SiO_2 capping layer

Cr or TaN absorber (~70 nm) with 50 nm Ru Buffer layer

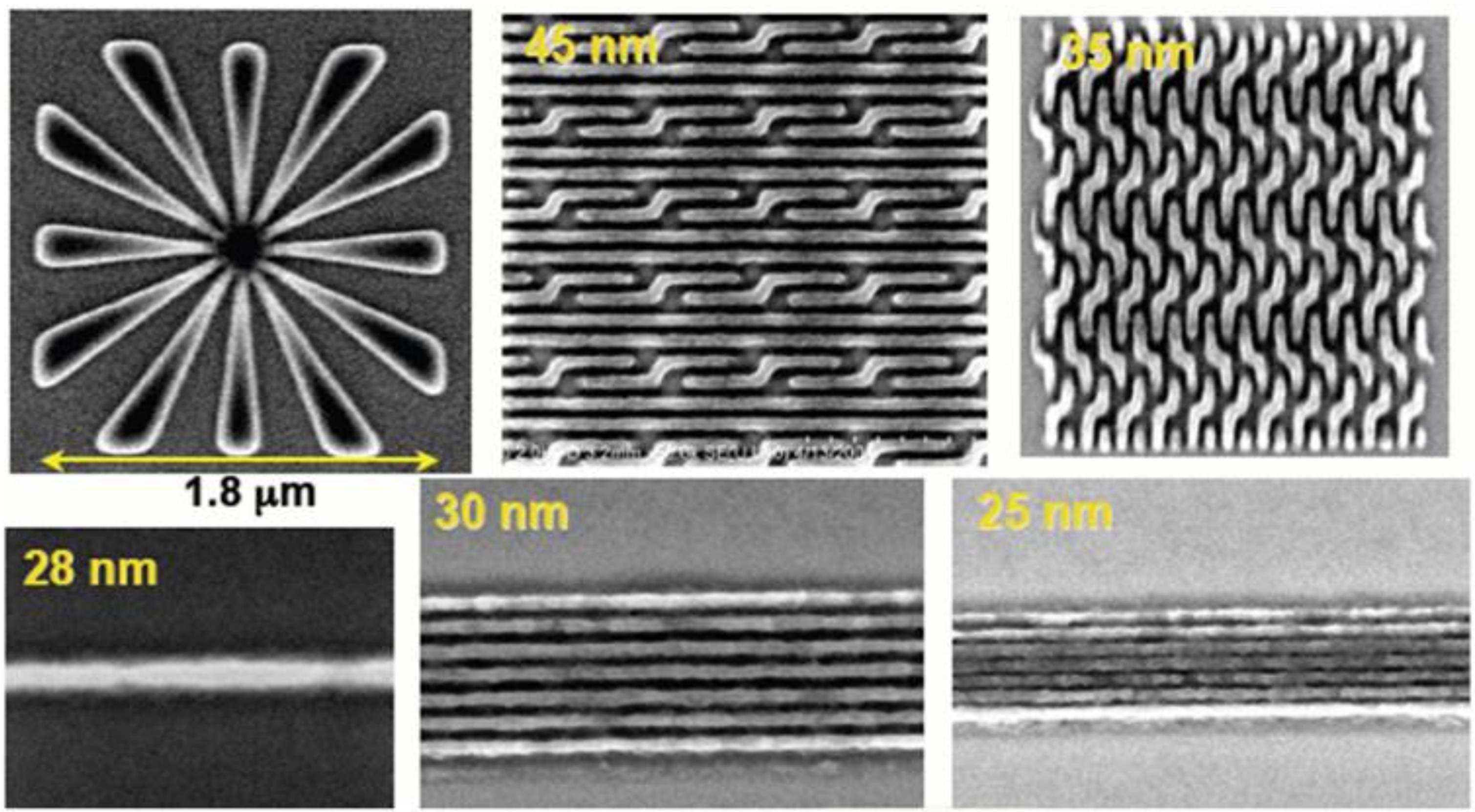
LTEM substrate (Ti-doped fused silica) ULE (Corning), or Zerodur (Schott)

Attribute	TaN	Cr	Comments
CD control	✓		TaN has smaller RIE CD bias
Cleaning			Both resistant to standard cleans
Emissivity		✓	
Inspection contrast	✓		TaN has higher contrast
Repair selectivity			Both need small improvement
Aspect ratio	✓		TaN can be 8 nm thinner than Cr

Step and Scan Exposure



EUV Resolution



LBNL

EUV Challenges

- Source power > 180 W at intermediate focus, acceptable utility requirements through increased conversion efficiency and sufficient lifetime of collector optics and source components
- Cost control and return on investment
- Resist with < 1.5 nm 3s LWR, < 10 mJ/cm2 sensitivity and < 20 nm ½ pitch resolution
- Fabrication of Zero Printing Defect Mask Blanks
- Establishing the EUVL mask Blank infrastructure (Substrate defect inspection, actinic blank inspection)
- Establishing the EUVL patterned mask infrastructure (Actinic mask inspection, EUV AIMs)
- Controlling optics contamination to achieve > five-year lifetime
- Protection of EUV masks from defects without pellicles
- Fabrication of optics with < 0.10 nm RMS figure error and < 7% intrinsic flare

ITRS 2009

Review（小 结）

- Photo-lithography and key factors（光刻机及性能参数）
- Photo resist and mask(掩膜与光刻胶）
- Lithography process（光刻工艺步骤）
- RET（提高分辨率的方法）
- EUV（新型光刻技术）

Homework

1. For immersion Lithography in different index fluids (NA = 0.93 for air), calculate the minimum width, respectively.

$n=1.63$, $\lambda = 193nm$, $k_1 = 0.25$,

$n=1$, $\lambda = 157nm$, $k_1 = 0.25$

Compare the results, and what conclusion could you draw?

(对于浸入式光刻，计算浸入上述不同液体介质时，光刻能达到的最小线宽。比较两种不同结果，你会得出什么样的结论或启示？)

2. The following picture shows DOF implications. In the figure, if DOF is smaller than △， what does it mean?

How can △ satisfy the requirements if clear photo images are required ?

(下图暗示了DOF的含义。图中如果△>DOF,意味着什么？若希望光刻得到清晰的图，则△必须满足什么条件？)